CM1 Bot Integrated Program Report

PETW Step 1, figure-and-table integration of verified CM1 evidence

Report date: 2026-08-12 | Evidence type: local audit/display derivative | Scientific execution by this report: none | Remote/HPC/solver/Zotero/LitWiki/source-archive modification: none

Current 40-case evidence	All-102 campaign	Scientific claim boundary
ACCEPT_IMPROVED_NOT_EXACT **24 / 40 Level-1 valid label**	**identity and release inventory** **102 unique, 61 release-ready new, 1 blocked**	**NOT_ESTABLISHED / NOT_SUPPORTED / NOT_PROMOTED** **formal 3-D validation and Vhat_3s unavailable**

使用边界。 本报告是 CM1 bot-level audit/display derivative。它整合已经存在的组件报告、图形、表格、review/Wiki inventory、运行状态和接口证据，不产生新的 solver 结果，不重新解释原始 NetCDF，不改变既有标签，也不提升 scientific claim。所有 figure 和 table 均保留来源路径、证据类别和 claim limit。

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1. Scope and evidence control

本报告回答 CM1 bot 已经形成了哪些可追溯证据，哪些证据属于模型内部结果、工程准备、review/Wiki 资产、停止或失败状态、display derivative，以及这些对象能否进入 Events、Windfield、CFD、EVA 和 Manuscript handoff。报告使用的控制文件是 CM1 bot review baseline、CM1 review family inventory、完成 bot-work audit、CM1 V4 primary-results package、BESSIE Job 3388245 V2 display package、D420 V12 display package、CM1 Path C STATE.md 和 PETW Master State。

证据层级保持分离。输入 identity、protocol、package、manifest 和 release inventory 是输入或工程准备证据。scheduler 和 application terminal records 是运行状态证据。CM1 diagnostic、M0/M1 fields 和后处理图是模型或 solver-derived evidence。review/Wiki archive PI approved claim 是独立的审批层，本报告没有执行该层。

报告范围保护。 本地报告目录只保存图形、表格、source packet 副本、HTML/PDF 和 QA receipts。没有复制 raw NetCDF、原始 sounding、远程源文件、凭据、token 或 scheduler 控制文件。

2. Bot characteristics and interfaces

2.1 Scientific and automation role

CM1 bot 是 PETW Step 1 的 mechanism-aware meteorological simulation and benchmark-reuse lane。它将 Events 的 event reconstruction packet 转换为受控模拟路线，恢复 Moore、Schumacher 等 benchmark 约束，固定 archive-member identity、sounding hash、namelist、executable、MPI decomposition、output writer 和 postprocessing version，并在输出侧区分 scheduler completion、CM1 application completion、model-internal diagnostics、Level-1 label、formal three-dimensional validation、Vhat_3s readiness 和 engineering use。

CM1 bot 的可审计自动化回路是：`input packet -> route and identity gate -> versioned run package -> runtime/output QA -> diagnostics and Level-1 adjudication -> source or handoff packet -> approval boundary`。现有证据主要处于 A0 planning/schema、A1 read-only ingestion 和 A2 supervised processing/QA。报告中没有把 package existence 或 QA pass 写成科学验证。

2.2 Interfaces

接口	输入或输出	CM1 返回	边界
Events -> CM1	parent storm mode, local mechanism hypotheses, confidence, detectability, station/radar context, surface signals, sounding availability, ambiguity reason, simulation_route_hint	case identity, protocol/input hashes, runtime identity, output manifest, model-internal mechanism diagnostics, near-surface response, footprint/duration/trajectory and uncertainty reasons	模型初始化不能覆盖观测证据，CM1 反馈不能自动提升 EVT、Path C 或 validated event。
CM1 -> Events	model-internal ascent, precipitation, descent, divergence, cold-pool, outflow and detection-limit findings	bounded diagnostic and failure/uncertainty packet	诊断不能被写成观测真值或事件机制真值。
CM1 <-> ComputeOps	CM1 owns scientific package, resource shape, output contract and interpretation; ComputeOps owns connection, staging, scheduler, monitoring, cancellation and terminal retrieval	verified operational receipts and terminal state	CM1 不输入密码、token 或 MFA，不绕过 ComputeOps 进行远程执行。
CM1 -> Windfield	variable semantics, cadence, spatial support, coordinates, source identity, finite-value checks and output manifest	wind-field output manifest only after semantics and claim boundary close	sws、diagnostic u10/v10 不能直接称为 engineering gust。
CM1 -> CFD	explicitly selected dense source packet, source time, grid, native variables, CM1 initial field, boundary drive, clock mapping and raw hash	CM1-to-CFD source identity and interpretation packet	CFD owns case, mesh, boundary dictionaries, solver and runtime QA; CM1 Level-1 label is not CFD authorization。
CM1 <-> Literature/LitWiki	benchmark methods, sounding, success definition and model-output semantics	frozen simulation protocol and source-semantic notes	文献 benchmark 或 retrieved claim 不能自动成为 PETW event truth 或 engineering design claim。

3. Wiki, RAG and review inventory

当前 CM1 scholarly asset state 不是一个 review 和一个 Wiki，而是两条 review manuscript family、多个版本化 PDF、两套成熟 governed Wiki/RAG namespace，以及一个从 Chinese v2.1 派生但尚未完成的 English counterpart。两个 review family 的 scientific approval 均未授予。

3.1 Review manuscript families

family	current asset	verification	status	limit
Family 1: Step 1 CM1 downburst modeling review	Chinese v2, 29 pages; English counterpart, 43 pages	English counterpart: 45/45 cited keys, two deterministic builds, 43-page visual QA and seven-file archive hash read-back	view; not_scientifically_approved	Does not establish completed PETW downburst reproduction, validated Moore benchmark, universal sounding, optimal grid, Vhat_3s, design value, structural load, or Path C validation.
Family 2: CM1/WRF downburst to high-rise building aerodynamic loads review	Chinese v2.1 CFD-focused revision, 26 pages; English v1.1 is a prior internal candidate	Chinese package retained 57 sources from 68 frozen sources and one bounded standalone-experiment edge VE-001	RNAL_REVIEW; not_scientifically_approved	VE-001 does not validate moving downbursts, floor forces, base moments, torsion, other aspect ratios, universal CFD error bounds, or the complete CM1/WRF-to-load chain.
English counterpart of Family 2 Chinese v2.1	No manuscript_en.tex or manuscript_en.pdf	Exact Stage 2 route failed before execution with 401 invalid_refresh_token/token_expired	ARCHIVED_CANDIDATE_BLOCKED; not_scientifically_approved	The missing counterpart is a production gap, not evidence for a stronger scientific claim.

3.2 Governed Wiki/RAG namespaces

namespace	corpus	Wiki/RAG	evaluation	boundary
cm1_to_cfd39_topic	39 canonical sources	54 active pages; 100 chunks, 100 claims, 24 relations	47/47 PASS	Evidence and retrieval substrate only; no Path C validation, design guidance or claim approval.
step1_cm1	45 source records	45/45 source-semantic verification	10/10 historical retrieval evaluation; complete_semantic_source_fidelity	review required; not_approved and not scientifically approved.

Review and Wiki inventory sources are copied under `source_packets/governing/` and cited in `EVIDENCE_MATRIX.csv`. Compilation, retrieval and archive acceptance remain distinct from scientific approval.

4. Work-state matrix

下表把当前 CM1 状态按 work item、证据对象和 claim ceiling 集中列出。详细 machine-readable version 位于 `WORK_STATE_MATRIX.csv`。

work_item	status	what_exists	source_reference	claim_limit
CM1 bot architecture	established	mechanism-aware meteorological simulation and benchmark-reuse lane; interfaces recorded	CM1 bot review baseline	No independent claim approval; A0-A2 supervised automation boundary.
Schumacher technical benchmark	runtime/scaling evidence established; full scientific benchmark NOT_ESTABLISHED	build, MPI, Thompson startup, NetCDF, 240 x 240 scaling and 480 x 480 output-writer diagnosis	CM1 Path C STATE.md and baseline	Do not call the full 1200 x 1200 benchmark complete.
Moore protocol and 40-case registry	ACCEPT_IMPROVED_NOT_EXACT	40 protocol-consistent reconstructed cases, bounded 2-D Level-1 model-internal screening	CM1 V4 registry and report	paper_97_membership=unknown; formal_three_dimensional_downburst_validation=false; Vhat_3s_available=false.
Moore 40-case labels	24 valid, 2 strong-needs-3D, 2 boundary, 9 weak, 3 scientifically invalid	registry-derived Level-1 counts; sws_max >= 26 m/s proxy 15/40 and 12/35 after stated exclusions	TABLE_40_DISTRIBUTION.csv and TABLE_40_CROSSTAB.csv	Labels are not observations, paper-pass labels, validated 10 m 3 s gusts, or Vhat_3s.
Moore all-102 identity audit	102/102 unique; 101 release-ready; 61 new release inventory; 1 blocked	input identity and release inventory only	TABLE_ALL102_STATUS.json and TABLE_ALL102_IDENTITY_STATUS.csv	SOURCE_HEIGHT_ORDER_BLOCKED case remains unmodified; no new all-102 scientific result.
Job 3396402	LIVE_EXECUTION, operational-only, scientific_result=false	current gate submission/status evidence retained as operational state	CM1 V4 sidecar and table	Not a scientific result and not evidence that all-102 cases completed.
BESSIE Job 3388245	completed traceable conditional output; Level-1 NOT_SUPPORTED under frozen gate	3600 s, 241 NetCDF files, 60 s stats cadence, spatial outputs at 0/900/1800/2700/3600 s	BESSIE V2 display and diagnostics	Downburst-specific reconstruction NOT_ESTABLISHED; validated 10 m wind NONE; Vhat_3s NONE.
D420 CM1 source Job 3379790	COMPLETED 0:0; source product	1026/1026 manifest, 20/20 cleanroom, 261/261 assembly, 261/261 frame-flux QA	D420 T1/T2 tables and report	CM1 source product is not CFD solver output.
D420 one-frame pilot	PILOT_PASS; cfd_handoff_allowed=false	single-frame native-to-centered volume and interface checks	D420 V12 report and T1/T3 tables	No multitime propagation, CFD physics, building pressure or structural response.
D420 M0 and M1	historical actual solver execution evidence	M0 Job 3384063 and M1 Job 3386652 have bounded execution and field QA evidence	D420 T2/T6/T7 tables	Not validation, not M1 superiority, not design guidance.
D420 Job 3393320	W; NOT_CONSISTENT_CONTROLLED_COMPARISON	four-time M0/M1 postprocessing ranges and figure inputs	D420 T2/T7 and boundary audit	Cannot attribute differences to one mesh, closure, wall, domain, initial state or boundary factor.
D420 matched-drive package	NOT_EXECUTED, generated preparation	100 m/10 m resolution and 10 m laminar/URANS closure axes	D420 T6 and V12 sidecar	No new solver result, propagation validation, scientific acceptance or approved claim.

5. Integrated evidence narrative

5.1 Frozen Moore protocol and input identity

CM1 V4 preserves `PETW_MOORE_UNIFORM_AGL_V1`, CM1 r20.3, an 800 x 800 x 84 grid, 7200 s integration, 60 s output cadence, output_filetype=3, 48 MPI ranks and 6 x 8 decomposition. The height contract is `z_i = HGHT_i - HGHT_surface`, with surface AGL set to 0 and pressure/profile fields preserved. This is an input contract and identity result. It does not establish paper membership or scientific completion.

5.2 Existing 40-case reconstructed ensemble

已有 40 个 protocol-consistent reconstructed production cases。Registry-derived Level-1 counts are 24 `VALID_DOWNBURST_CANDIDATE_LEVEL1`, 2 `STRONG_OUTFLOW_CANDIDATE_NEEDS_3D`, 2 `BOUNDARY_TRUNCATED`, 9 `WEAK_NEAR_SURFACE_OUTFLOW` and 3 `COMPLETED_BUT_SCIENTIFICALLY_INVALID`。独立的 `sws_max >= 26 m/s` 是 Moore-style strength/component proxy, not Vhat_3s and not a validated 10 m 3 s gust. The current benchmark status is `ACCEPT_IMPROVED_NOT_EXACT`; `paper_97_membership=unknown`.

level1_label	count
VALID_DOWNBURST_CANDIDATE_LEVEL1	24

level1_label	count
WEAK_NEAR_SURFACE_OUTFLOW	9
COMPLETED_BUT_SCIENTIFICALLY_INVALID	3
BOUNDARY_TRUNCATED	2
STRONG_OUTFLOW_CANDIDATE_NEEDS_3D	2

5.3 All-102 identity and operational status

完整 denominator 是 102 archive identities。102/102 identity unique，现有生产 40，remaining 62，其中 61 个 release-ready new inputs 和 1 个 `SOURCE_HEIGHT_ORDER_BLOCKED` case。blocked case `20190630_2100_42.870_-91.890` 的原始高度顺序导致转换后 0.00 m AGL 后接 -1.76 m AGL；没有排序、删除、clamp、插值或 relabel。Job `3396402` 保持 `LIVE_EXECUTION`、`scientific_result=false`、operational-only。该记录不能表示 102 个 scientific cases 完成。

field	value
archive_identities	102
existing_40_production	40
remaining	62
release_ready_new	61
source_height_blocked	1
blocked_case	20190630_2100_42.870_-91.890
paper_97_membership	unknown
job_id	3396402
job_status	LIVE_EXECUTION
scientific_result	False

5.4 BESSIE conditional CM1 output

BESSIE Job `3388245` 是已完成、可追溯的 conditional CM1 output。V2 package records 241 NetCDF files, 48 ranks per spatial time, spatial output at 0, 900, 1800, 2700 and 3600 s, and stats cadence 60 s. Existing model-internal sequence is ascent, precipitation, descent, local divergence and cold-pool diagnostics. Source temporal extrema are summarized below.

诊断	峰值	时间	解释边界
wmax	22.4418 m/s	2340 s	max vertical velocity, model-internal
pratemax	0.00936125 kg m-2 s-1	2400 s	precipitation-rate diagnostic
wmin	-11.4658 m/s	2520 s	min vertical velocity, model-internal
divmax	0.00118535 s-1	2580 s	local divergence diagnostic
s10 max	approximately 2.98344 m/s	source temporal sequence	s10 remains distinct from lowest-model-level wind and u10/v10

BESSIE frozen gate。 `downburst-specific reconstruction: NOT_ESTABLISHED`; `VALID_DOWNBURST_CANDIDATE_LEVEL1: NOT_SUPPORTED`; `Path C: NOT_PROMOTED`; validated 10 m wind: NONE; `Vhat_3s: NONE`. The spatial snapshots are discrete output observations, not a continuous trajectory.

5.5 D420 CM1-to-CFD evidence

D420 CM1 source-production Job `3379790` completed with source-package manifest 1026/1026 PASS, cleanroom tests 20/20 PASS, assembly QA 261/261 PASS and frame-flux QA 261/261 PASS. The one-frame velocity-volume pilot independently checked native staggered u/v/w, centered representation, coordinate destaggering, duplicate ownership and NetCDF writing. It remains an interface pilot, with `cfd_handoff_allowed=false`, not a multitime propagation result.

Historical M0 Job `3384063` and M1 Job `3386652` retain actual solver execution and field QA evidence. M0 F7/F8 are existing solver-derived postprocessing displays; M1 is restart execution and field completeness evidence. Job `3393320` generated bounded M0/M1 exploratory metrics at selected times without a new solver execution. The original boundary audit must remain `NOT_CONSISTENT_CONTROLLED_COMPARISON` because wall treatment, closure, domain, initial/restart state, outer-boundary fallback, spatial support and write interval differ. Jobs `3389204` and `3391286` failed before scientific comparison products and are retained as stopped execution evidence, not results.

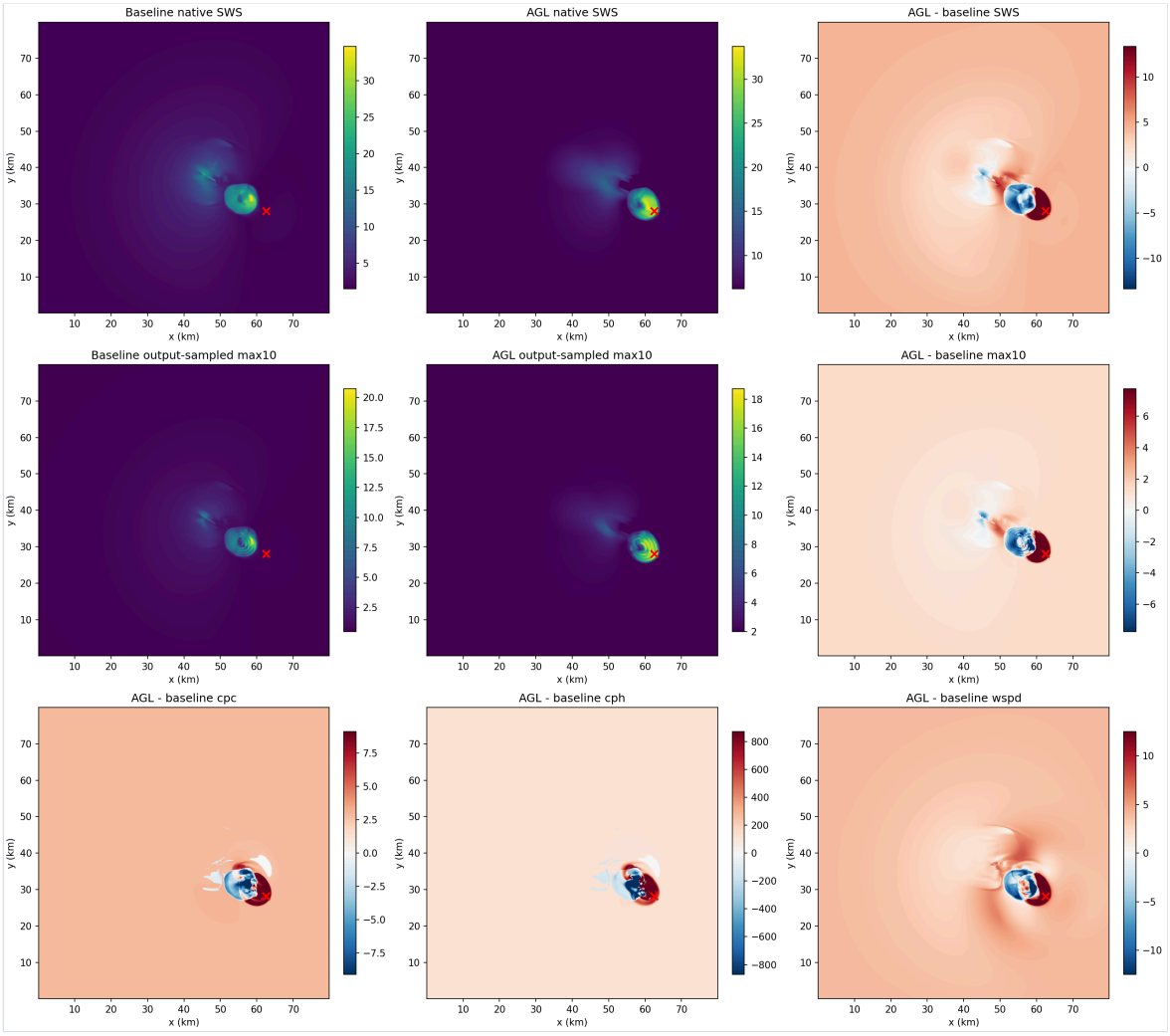
5.6 Failure and stop semantics

CM1 evidence keeps the earliest failure or gate reason. A package validator failure, missing processor mesh points, a source-height ordering blocker, a frozen BESSIE Level-1 gate, or a missing English review counterpart is not relabeled as a scientific negative or silently repaired. This preserves the difference between failed execution, stopped execution, incomplete package, blocked input, model-internal diagnostic, display derivative and scientific interpretation.

6. Integrated figures

本节包含全部 12 张 CM1 V4 figures、7 张 BESSIE V2 figures 和 8 张 D420 V12 figures，共 27 个 report-local image assets。每张图均有实际 source path、变量和单位、时间或坐标语义、证据类别、source hash、copy hash 和 claim limit。文件级索引位于 `FIGURE_INDEX.csv` 和 `FIGURE_INDEX.md`。

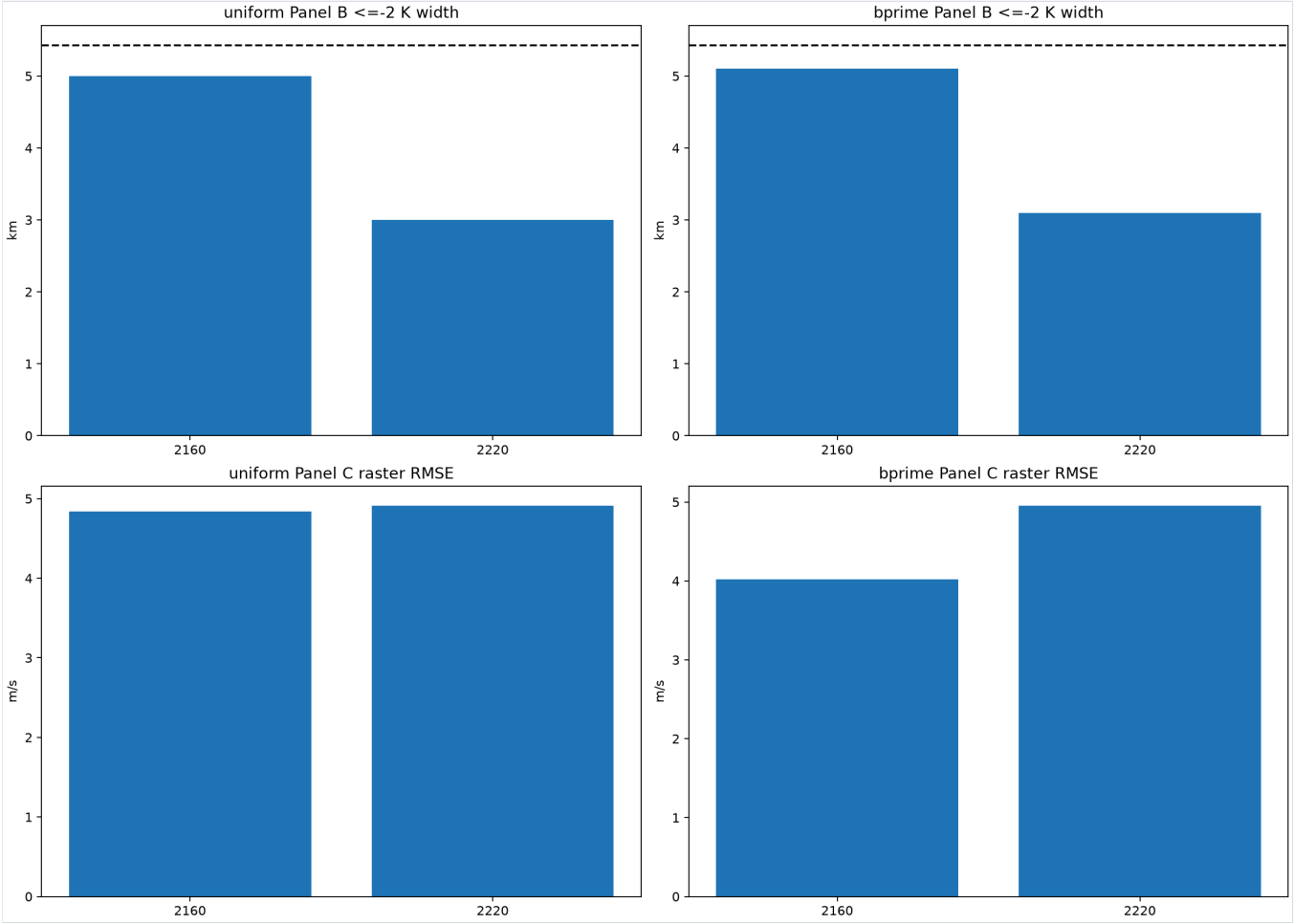
IF-01 **CM1 V4 primary display, F01**



内容。 Three-by-three spatial comparison: baseline native SWS, AGL native SWS, AGL minus baseline SWS, output-sampled max10, CPC/CPH/WSPD differences; red author-point marker where present.
变量和单位。 SWS, max10, CPC, CPH, WSPD and differences
时间和坐标。 Benchmark comparison asset; associated 2160/2220 s comparisons are explicit in companion metrics asset; axes and units: x/y in km; field-specific color bars retain source units
证据类别。 `source-traceable display derivative`
来源。

claim limit。 Display evidence only; does not establish exact Moore 97 membership, observational truth, formal 3-D validation, Vhat_3s, validated 10 m 3 s gust, design wind/load, or CFD solver validation.
源 SHA-256。 `8d792ada678bba8e662cf2f73dad200c2a8bd9a3cf65a40ad58577cea3699eda`；**报告副本 SHA-256。** `8d792ada678bba8e662cf2f73dad200c2a8bd9a3cf65a40ad58577cea3699eda`；**副本变换。** `none`

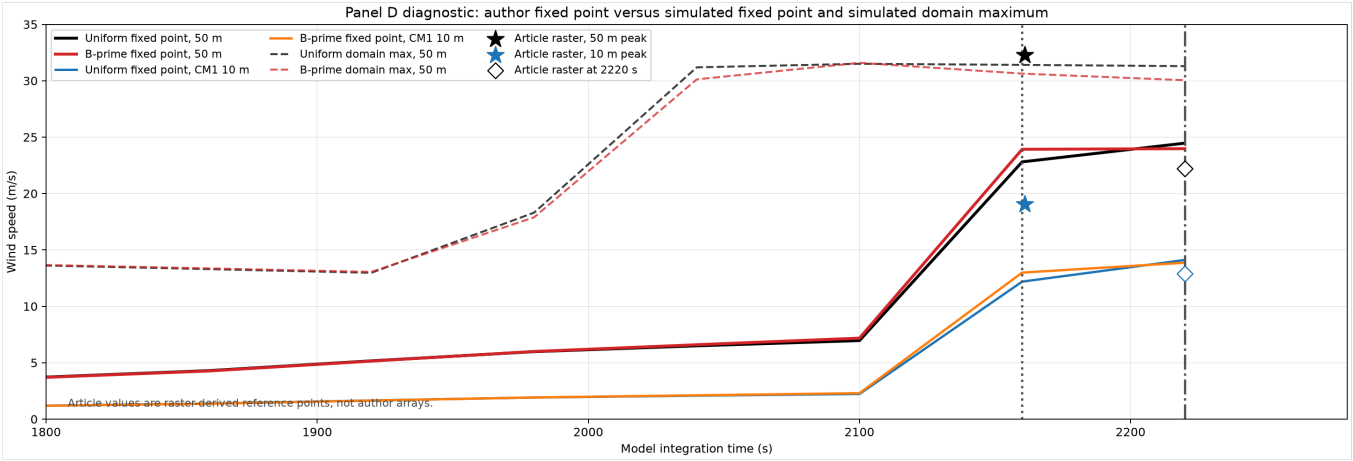
IF-02 **CM1 V4 primary display, F02**



内容。 Four-panel metric comparison for uniform and B-prime cases, including Panel B cold-pool width and Panel C raster RMSE.
变量和单位。 Panel B cold-pool width; Panel C raster RMSE
时间和坐标。 2160 s and 2220 s; axes and units: width in km; RMSE in m/s
证据类别。 `source-traceable display derivative`
来源。

claim limit。 Display evidence only; does not establish exact Moore 97 membership, observational truth, formal 3-D validation, Vhat_3s, validated 10 m 3 s gust, design wind/load, or CFD solver validation.
源 SHA-256。 `da81a575bc2c9f50ab65414d3d5d5e4e4caeb070f453a0a0ef24300acd6b3be4` ; **报告副本 SHA-256。** `da81a575bc2c9f50ab65414d3d5d5e4e4caeb070f453a0a0ef24300acd6b3be4` ; **副本变换。** `none`

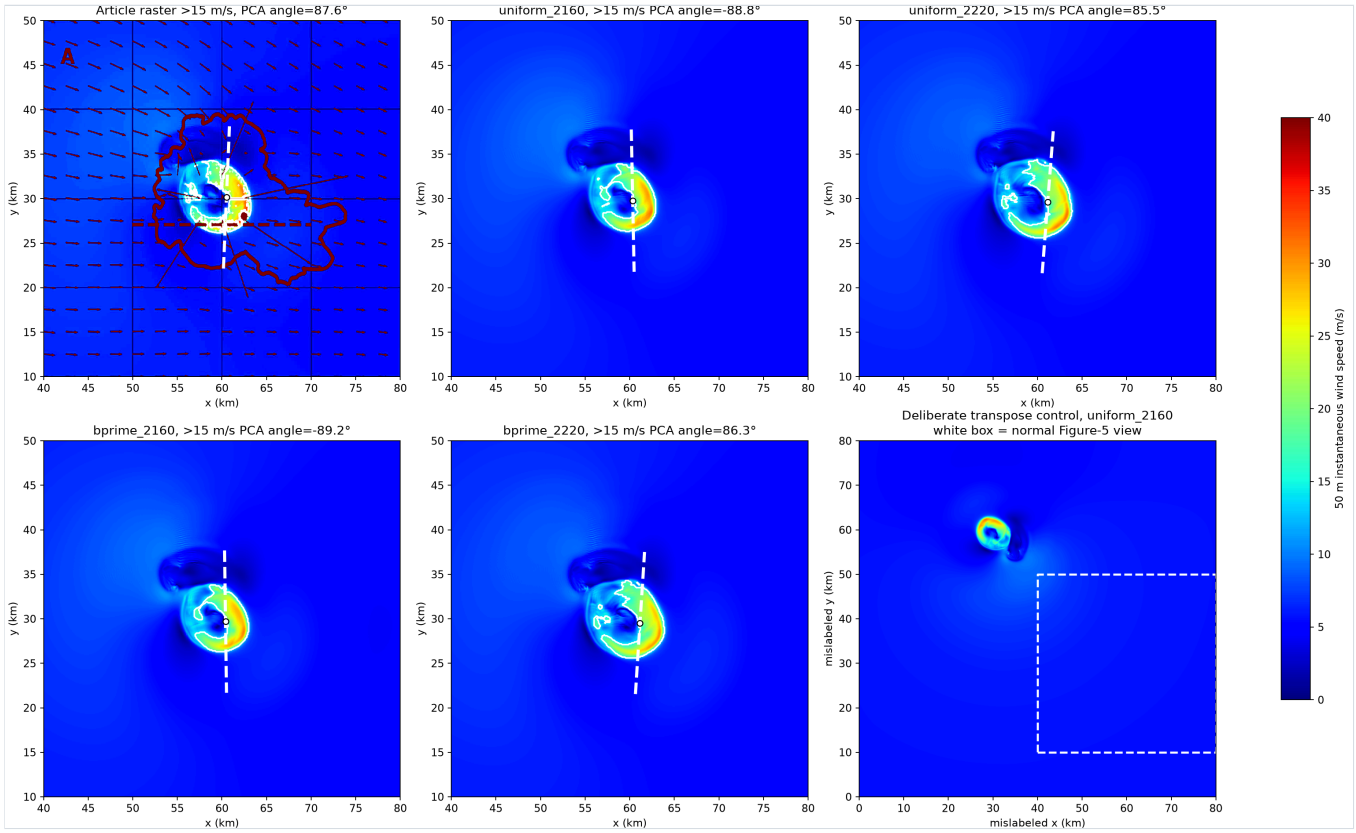
IF-03 **CM1 V4 primary display, F03**



内容。 Fixed-point versus domain-maximum diagnostic comparison with article-raster reference points; fixed-point and domain-max series are kept distinct.
变量和单位。 Wind-speed diagnostics, fixed-point values, domain maxima, article-raster reference points
时间和坐标。 Time series; raster values are reference points, not author arrays; axes and units: model integration time in s; wind speed in m/s
证据类别。 `source-traceable display derivative`
来源。

claim limit。 Display evidence only; does not establish exact Moore 97 membership, observational truth, formal 3-D validation, Vhat_3s, validated 10 m 3 s gust, design wind/load, or CFD solver validation.
源 SHA-256。 `7ff5416e1bac0ec04afbdbf16c4ffb7ed56bda7021006e49006bc32ca13cfb3c` ; **报告副本 SHA-256。** `7ff5416e1bac0ec04afbdbf16c4ffb7ed56bda7021006e49006bc32ca13cfb3c` ; **副本变换。** `none`

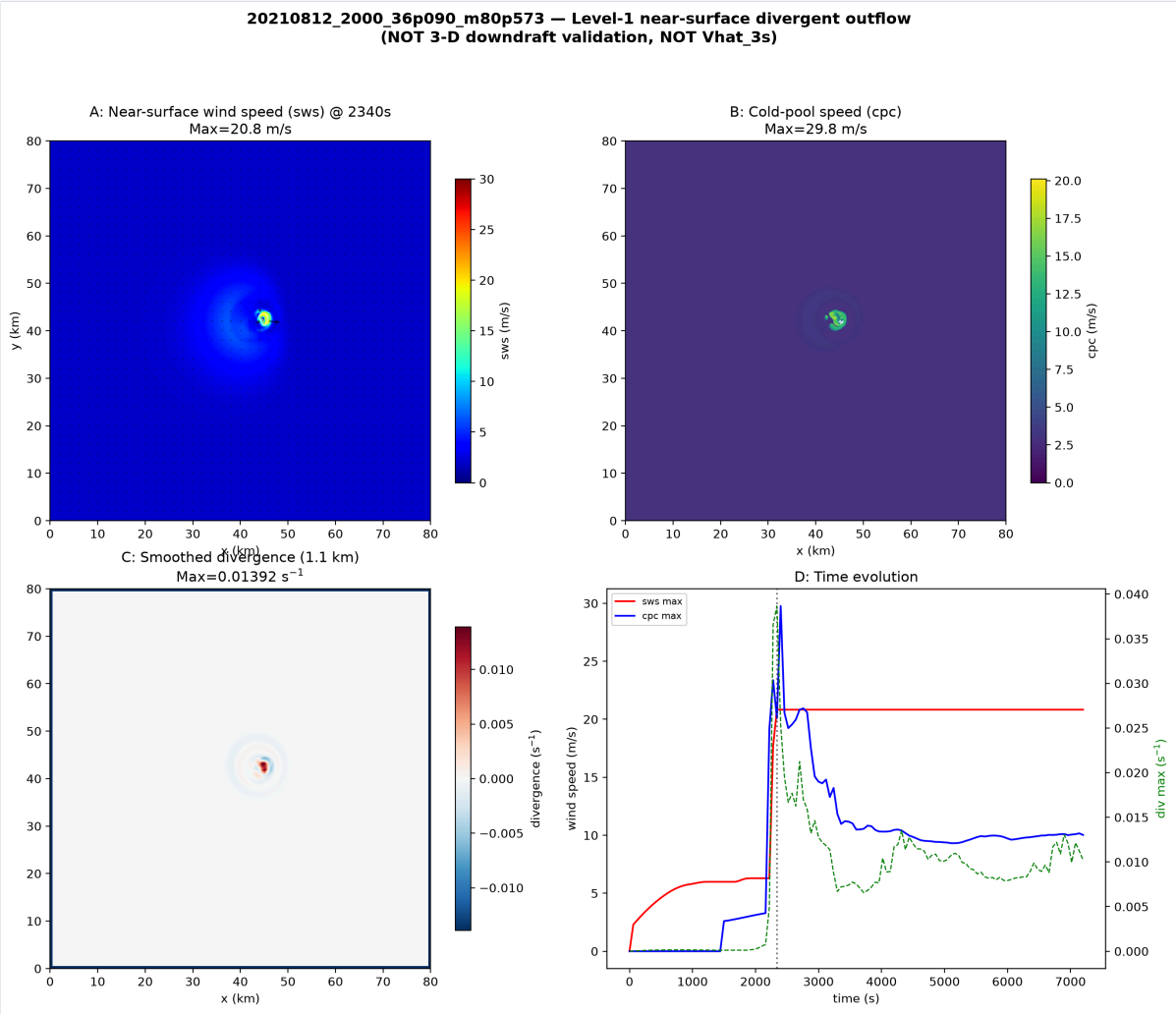
IF-04 **CM1 V4 primary display, F04**



内容。 Six-panel horizontal-orientation/PCA audit with article raster, uniform/B-prime 2160/2220 comparisons, and deliberate transpose control.
变量和单位。 Instantaneous 50 m wind speed, horizontal orientation/PCA fields
时间和坐标。 2160 s and 2220 s panels where labelled; transpose is a control, not a physical result; axes and units: x/y in km; instantaneous wind speed in m/s
证据类别。 `source-traceable display derivative`
来源。

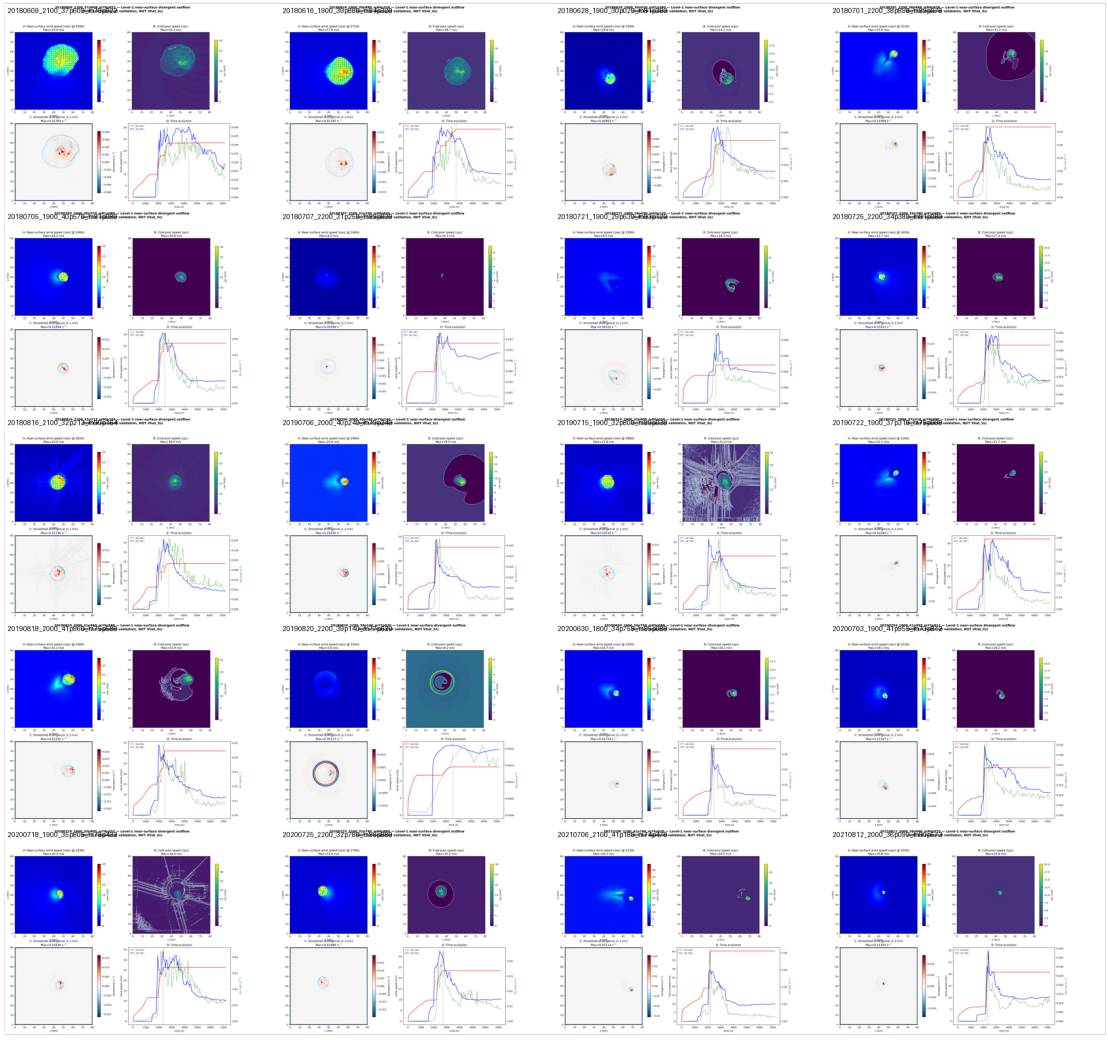
claim limit。 Display evidence only; does not establish exact Moore 97 membership, observational truth, formal 3-D validation, Vhat_3s, validated 10 m 3 s gust, design wind/load, or CFD solver validation.
源 SHA-256。 `21f9cd3c67ba9cf9bcb35eb1357e472a2f4fb682c353c9aa1df84c34339d31e9`；**报告副本 SHA-256。** `21f9cd3c67ba9cf9bcb35eb1357e472a2f4fb682c353c9aa1df84c34339d31e9`；**副本变换。** `none`

IF-05 **CM1 V4 primary display, F05**



内容。 Case 20210812_2000_36p090_m80p573, Level-1 near-surface divergent outflow. Four panels: near-surface SWS at 2340 s, cold-pool speed CPC, smoothed divergence at 1.1 km, and time evolution. Title preserves NOT 3-D downburst validation and NOT Vhat_3s.
变量和单位。 sws, cpc, smoothed divergence, time-evolution sws/cpc/divergence diagnostics
时间和坐标。 Panel A snapshot at 2340 s; Panel D extends through the 7200 s integration window; axes and units: x/y in km; wind speeds in m/s; divergence in s^-1; time in s
证据类别。 `source-traceable model-internal Level-1 display derivative`
来源。
claim limit。 Supports only bounded 2-D model-internal Level-1 evidence for the named case; not the author benchmark case, not formal 3-D validation, and not Vhat_3s.
源 SHA-256。 `df9c671c7cec936e304c50b27443b36c4e6955ff038035a1bfad33ff4f097351`；**报告副本 SHA-256。** `df9c671c7cec936e304c50b27443b36c4e6955ff038035a1bfad33ff4f097351`；**副本变换。** `none`

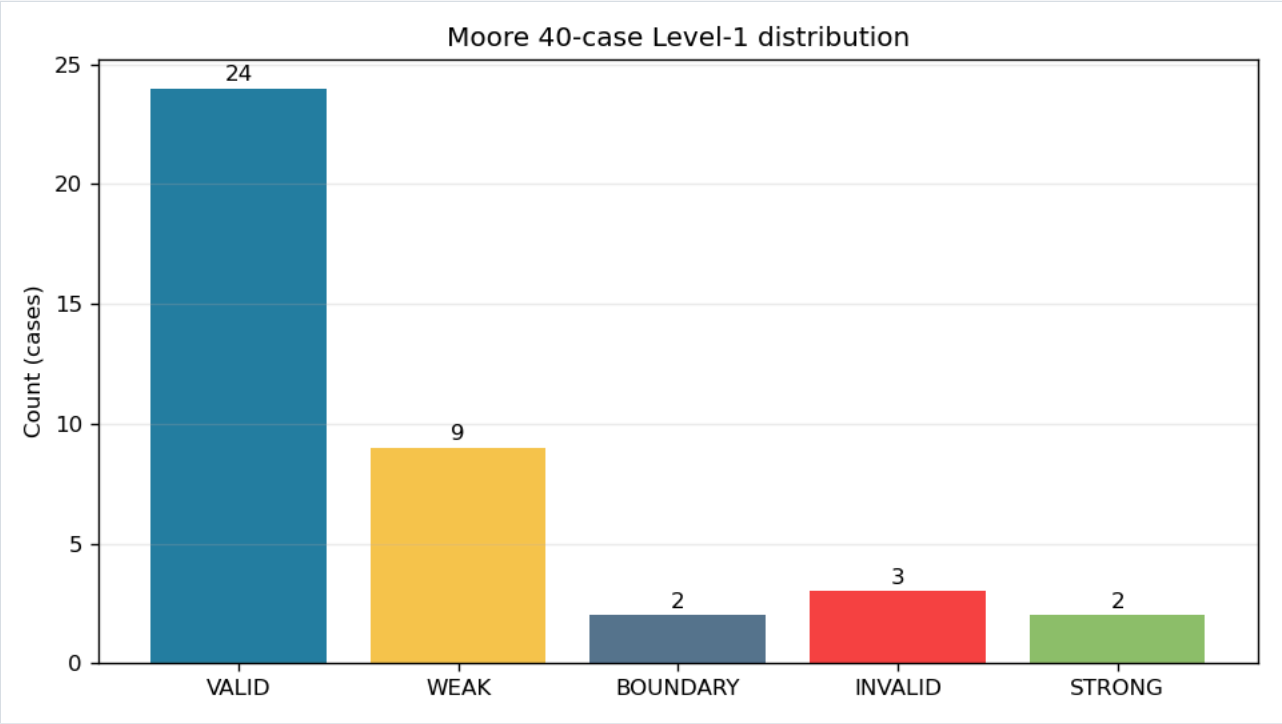
IF-06 **CM1 V4 primary display, F06**



内容。 Extension20 Figure-5-style contact sheet containing 20 visible case products, with repeated spatial fields, color bars, case/time labels, and time-evolution panels.
变量和单位。 Source Figure-5-style spatial and time-evolution diagnostics
时间和坐标。 Case/time labels are embedded per panel; this is a multi-case display, not a continuous aggregate time series; axes and units: Source panels retain their own km, m/s, and s units
证据类别。 `source-traceable display derivative`
来源。

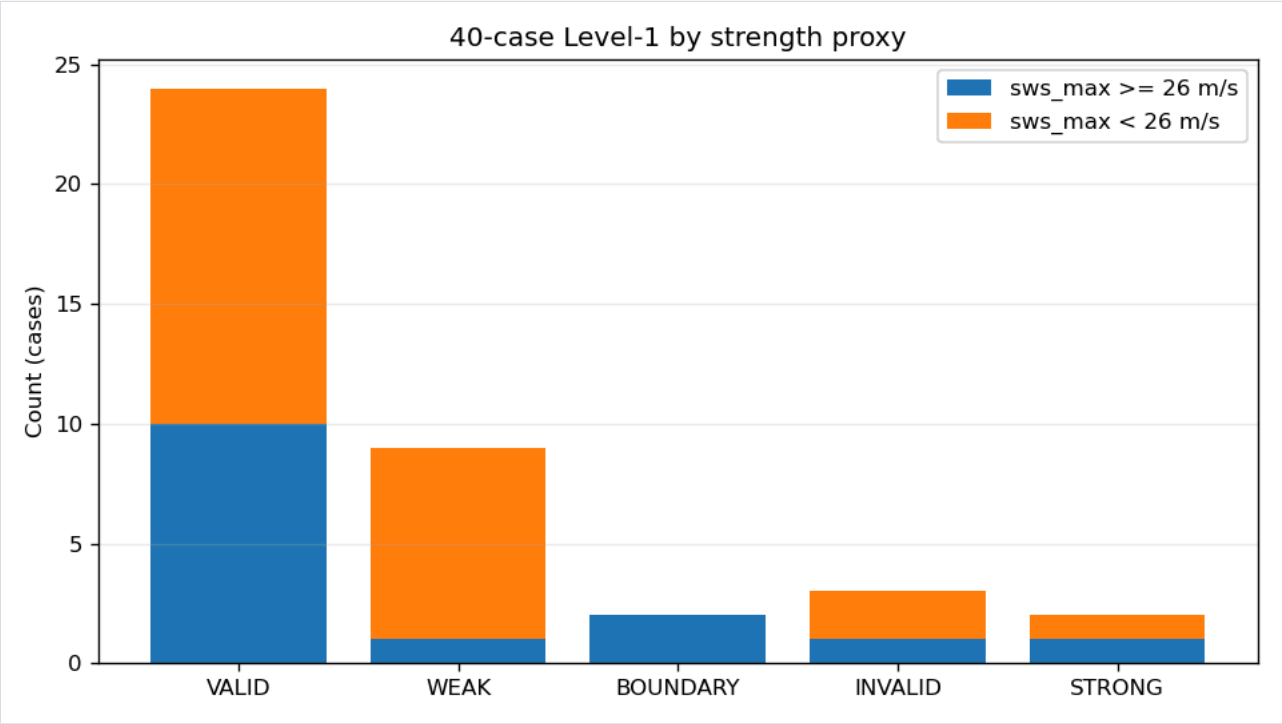
claim limit。 Supports display of 20 extension20 products only; does not prove all 40 cases, paper membership, or exact Moore reproduction.
源 SHA-256。 `f960ef9d80eaa6973482cfae4b0acd6d09ad8a78bcaa86cae58f8245ffee6623`；**报告副本 SHA-256。** `f960ef9d80eaa6973482cfae4b0acd6d09ad8a78bcaa86cae58f8245ffee6623`；**副本变换。** `none`

IF-07 **CM1 V4 primary display, F07**



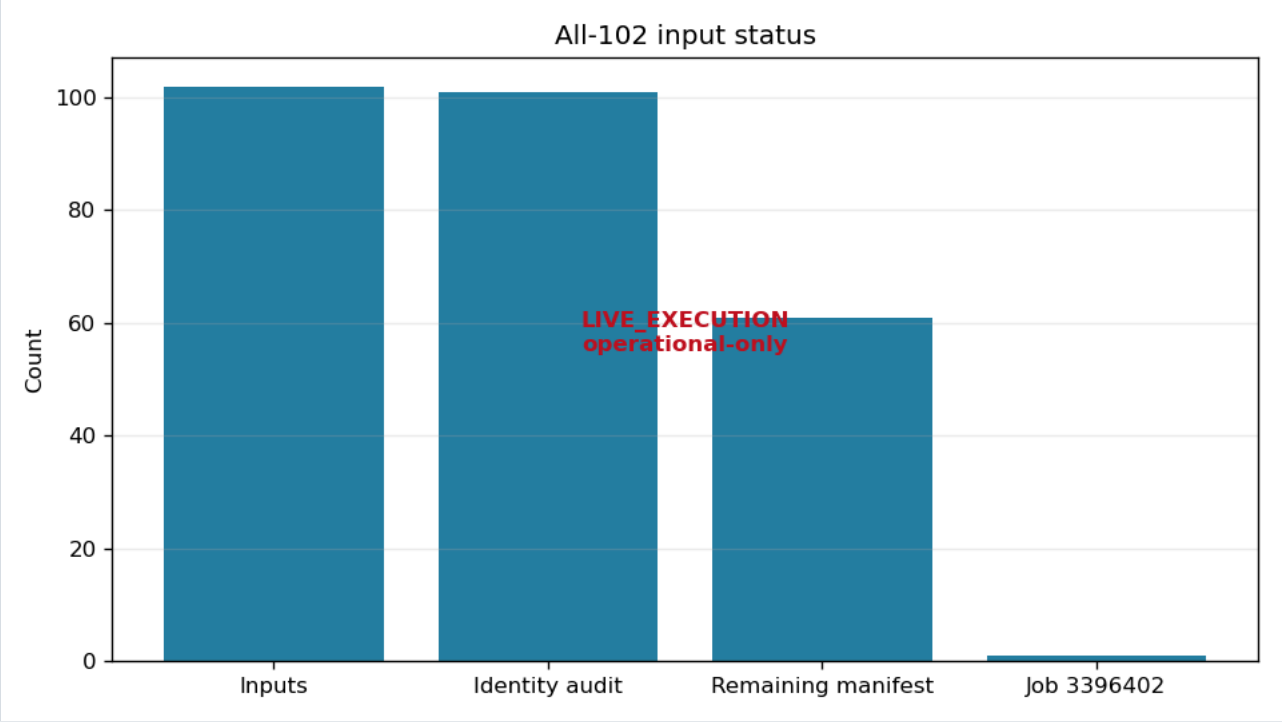
内容。 Moore 40-case Level-1 distribution with five registry-derived categories: VALID 24, WEAK 9, BOUNDARY 2, INVALID 3, STRONG 2.
变量和单位。 Level-1 category counts
时间和坐标。 Aggregate over the existing 40 completed two-hour cases; not a time series; axes and units: y axis: cases; x axis: Level-1 labels
证据类别。 `package-generated registry display derivative`
来源。
`plus moore40_level1_dual_axis_registry_v1.json`
claim limit。 Supports bounded 40-case model-internal screening counts only; labels are not observations, paper-pass labels, Vhat_3s, or validated gusts.
源 SHA-256。 `package-generated` ; **报告副本 SHA-256。** `7427191b981185831e0eb5d5765cef8b7648755a5ba0affac35152c95d13cdcc` ; **副本变换。** `none`

IF-08 **CM1 V4 primary display, F08**



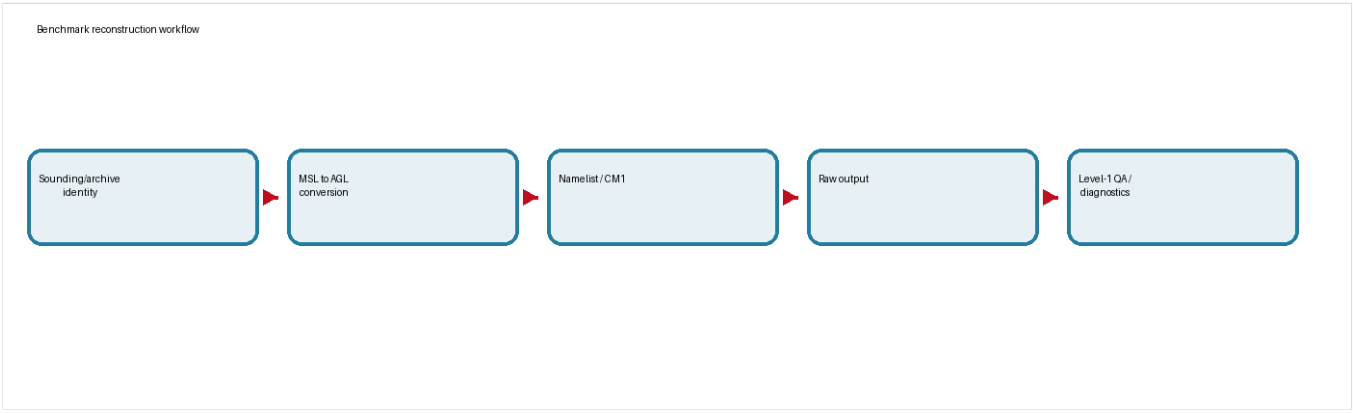
内容。 Stacked category comparison split by the screening proxy sws_max >= 26 m/s versus sws_max < 26 m/s.
变量和单位。 Level-1 category and cumulative lowest-model-level sws proxy
时间和坐标。 Aggregate over 40 cases; not a continuous physical time series; axes and units: y axis: cases; proxy threshold in m/s
证据类别。 `package-generated registry display derivative`
来源。 `plus`
`moore40_level1_dual_axis_registry_v1.json`
claim limit。 The 26 m/s axis is a Moore-style strength/component proxy, not Vhat_3s, not a validated 10 m 3 s gust, and not a paper-membership decision.
源 SHA-256。 `package-generated` ; **报告副本 SHA-256。** `85056c81ca721cc8a3ebae9a13d41a985bc3d246aeffda8cd3315d5458edf10b` ; **副本变换。** `none`

IF-09 **CM1 V4 primary display, F09**



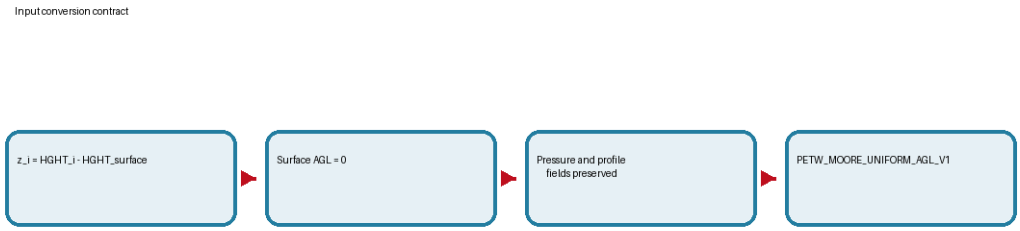
内容。 All-102 status display with Inputs 102, Identity audit 101, Remaining manifest 61, and Job 3396402 1, annotated LIVE_EXECUTION operational-only.
变量和单位。 Archive identity, release-inventory, and job-status counts
时间和坐标。 Current input/status snapshot; not a solver time series; axes and units: y axis: records/status items; x axis: status categories
证据类别。 `package-generated operational/status display derivative`
来源。
`plus moore102_input_identity_audit_v1.json and moore102_remaining61_input_release_manifest_v1.json`
claim limit。 Does not mean 102 scientific cases completed. Job 3396402 is operational-only and scientific_result=false.
源 SHA-256。 `package-generated` ; **报告副本 SHA-256。** `4de81736322f0627d57d9bdb011725a94dddaa5cff09908def13cbe8604c51d3` ; **副本变换。** `none`

IF-10 **CM1 V4 primary display, F10**



内容。 Protocol workflow diagram: Sounding/archive identity -> MSL to AGL conversion -> Namelist / CM1 -> Raw output -> Level-1 QA / diagnostics.
变量和单位。 Workflow stages and evidence classes
时间和坐标。 Protocol diagram, not a scientific time series; axes and units: No physical axes; protocol diagram
证据类别。 `package-generated protocol display derivative`
来源。
`and H-20260811-002_CM1_Primary_Results_Report_Request.md`
claim limit。 Documents process and evidence boundaries only; does not prove solver completion or scientific validation.
源 SHA-256。 `package-generated` ; **报告副本 SHA-256。** `6892ae859f09f473776681e98c295e0bdf1f71dba4f9cca3c90fcdb48bea91db` ; **副本变换。** `none`

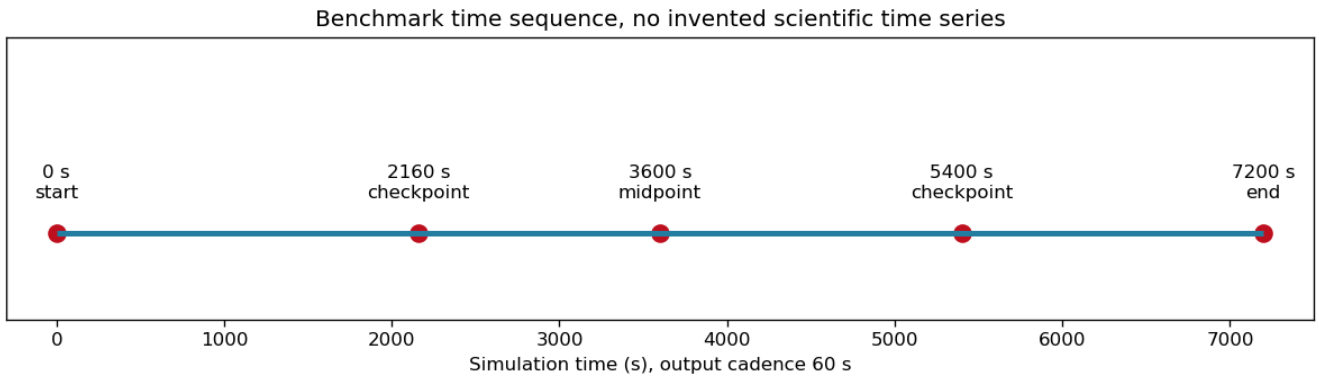
IF-11 **CM1 V4 primary display, F11**



内容。 Input conversion diagram showing z_i = HGHT_i - HGHT_surface, Surface AGL = 0, pressure/profile fields preserved, and PETW_MOORE_UNIFORM_AGL_V1.
变量和单位。 HGHT, AGL height, pressure/profile fields, protocol identity
时间和坐标。 Protocol diagram, not a scientific time series; axes and units: No physical plot axes; height conversion contract
证据类别。 `package-generated protocol display derivative`
来源。

`and moore102_input_identity_audit_v1.json`
claim limit。 Documents the frozen conversion contract; input identity does not prove paper membership or scientific completion.
源 SHA-256。 `package-generated` ; **报告副本 SHA-256。** `754b9f3d3ef247ce9e9773fbf3c768be7a5a21ce28a4e4e13df973facf762501` ; **副本变换。** `none`

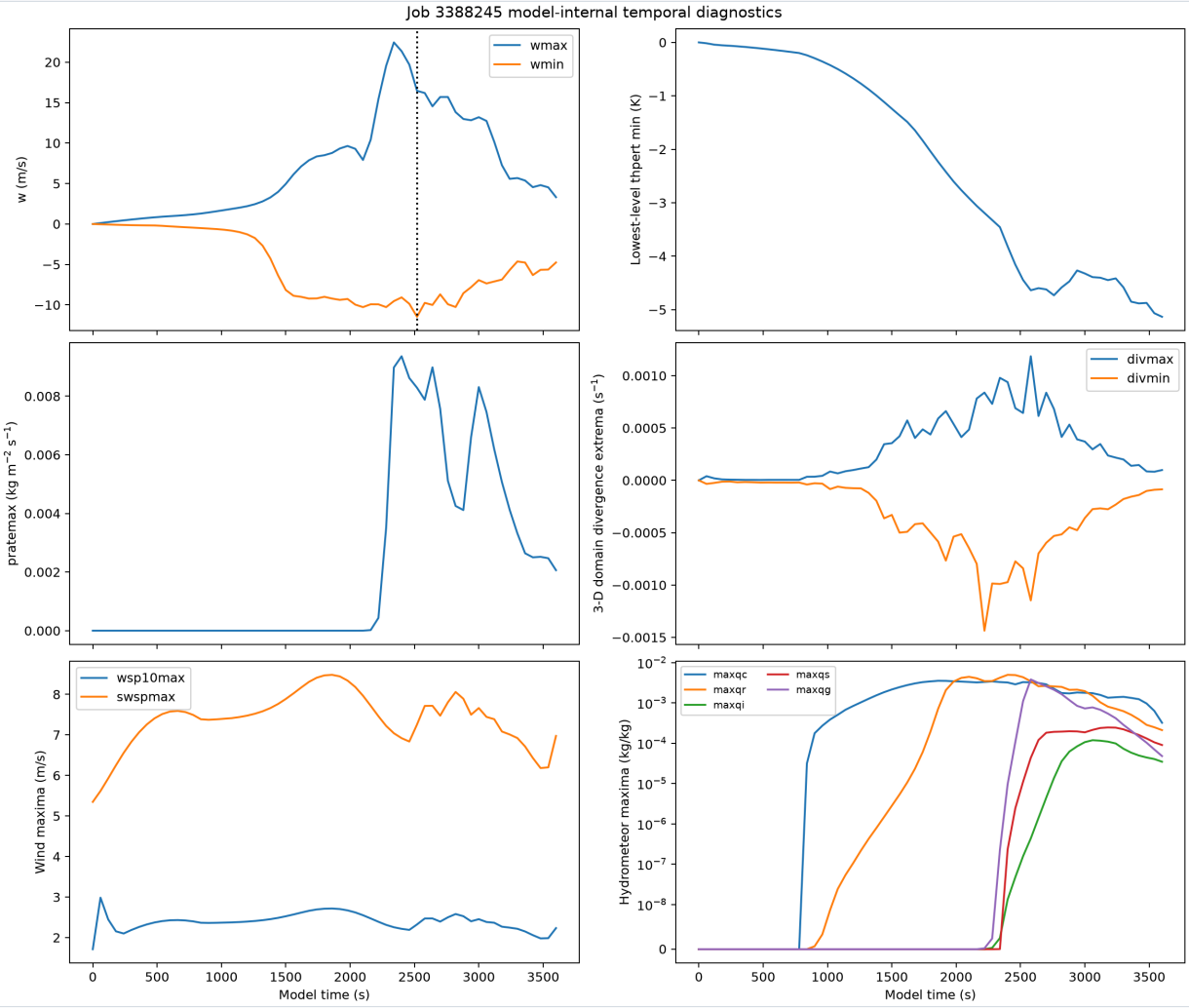
IF-12 **CM1 V4 primary display, F12**



内容。 Benchmark protocol timeline showing 0 s start, 2160 s checkpoint, 3600 s midpoint, 5400 s checkpoint, and 7200 s end.
变量和单位。 Simulation integration-time checkpoints and output cadence
时间和坐标。 0 to 7200 s protocol window with 60 s output cadence; axes and units: x axis: simulation time in s; cadence: 60 s
证据类别。 `package-generated protocol display derivative`
来源。

`and TABLE_BENCHMARK_PROCESS_TIMELINE.csv`
claim limit。 A checkpoint/cadence diagram, not an invented scientific time series and not evidence that every all-102 case completed.
源 SHA-256。 `package-generated` ; **报告副本 SHA-256。** `4ff7450bf7d04f7331640a0c9f48454f921cabaabb45d2acdd5b37e0c3186b4e` ; **副本变换。** `none`

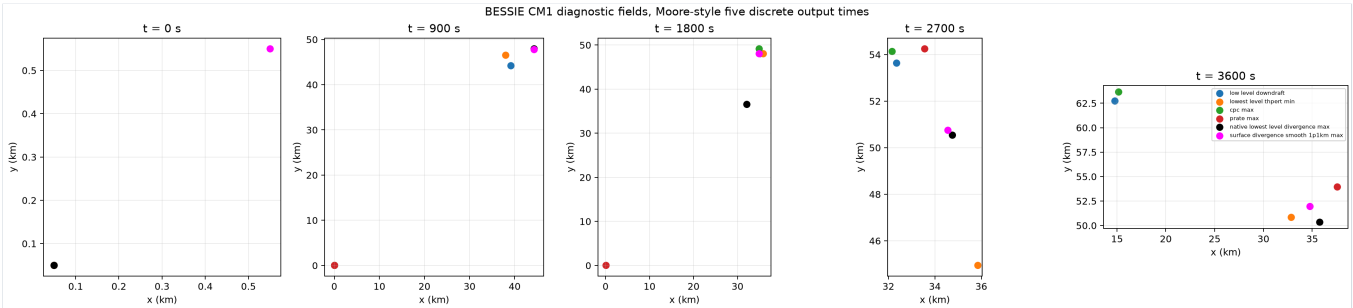
IF-13 BESSIE V2 display, F01



内容。 0 to 3600 s temporal diagnostics with source markers at 2340, 2400, 2520, and 2580 s; stats have no spatial position.
变量和单位。 wmax, wmin, precipitation rate, divergence, s10 and related source diagnostics; source units retained.
时间和坐标。 0 to 3600 s; stats cadence 60 s.
证据类别。 `source-traceable display derivative`
来源。

claim limit。 Does not establish event reconstruction, formal downburst validation, validated 10 m wind, Vhat_3s, design wind, or design load.
源 SHA-256。 `30bed722f1fe184f41586dc0977b492a52d0893b8b5c17c0dd516f997db2b30e`；**报告副本 SHA-256。** `30bed722f1fe184f41586dc0977b492a52d0893b8b5c17c0dd516f997db2b30e`；**副本变换。** `none`

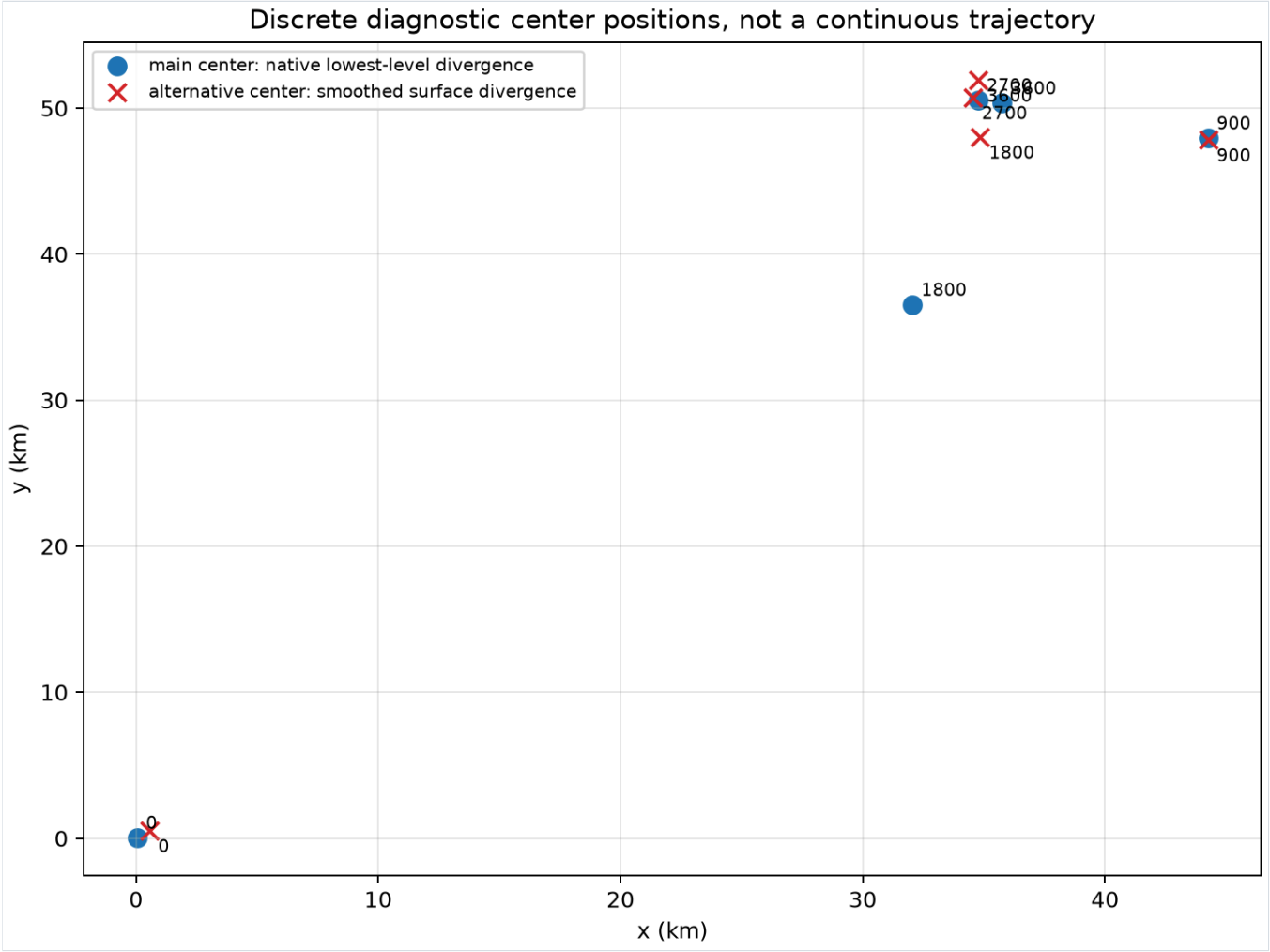
IF-14 BESSIE V2 display, F02



内容。 Five discrete diagnostic-center snapshots at exactly 0, 900, 1800, 2700, and 3600 s.
变量和单位。 Moore-style spatial diagnostics; source units retained.
时间和坐标。 Five observed output times only; no interpolation and no continuous trajectory.
证据类别。 `source-traceable display derivative`
来源。

claim limit。 The snapshots do not establish a continuous track or exclude a gust front or multi-feature process.
源 SHA-256。 `9fd5e4d031a2b88b9404e484126b647d8fbb798dc0dd33bc6f70f126bf559441`；**报告副本 SHA-256。** `9fd5e4d031a2b88b9404e484126b647d8fbb798dc0dd33bc6f70f126bf559441`；**副本变换。** `none`

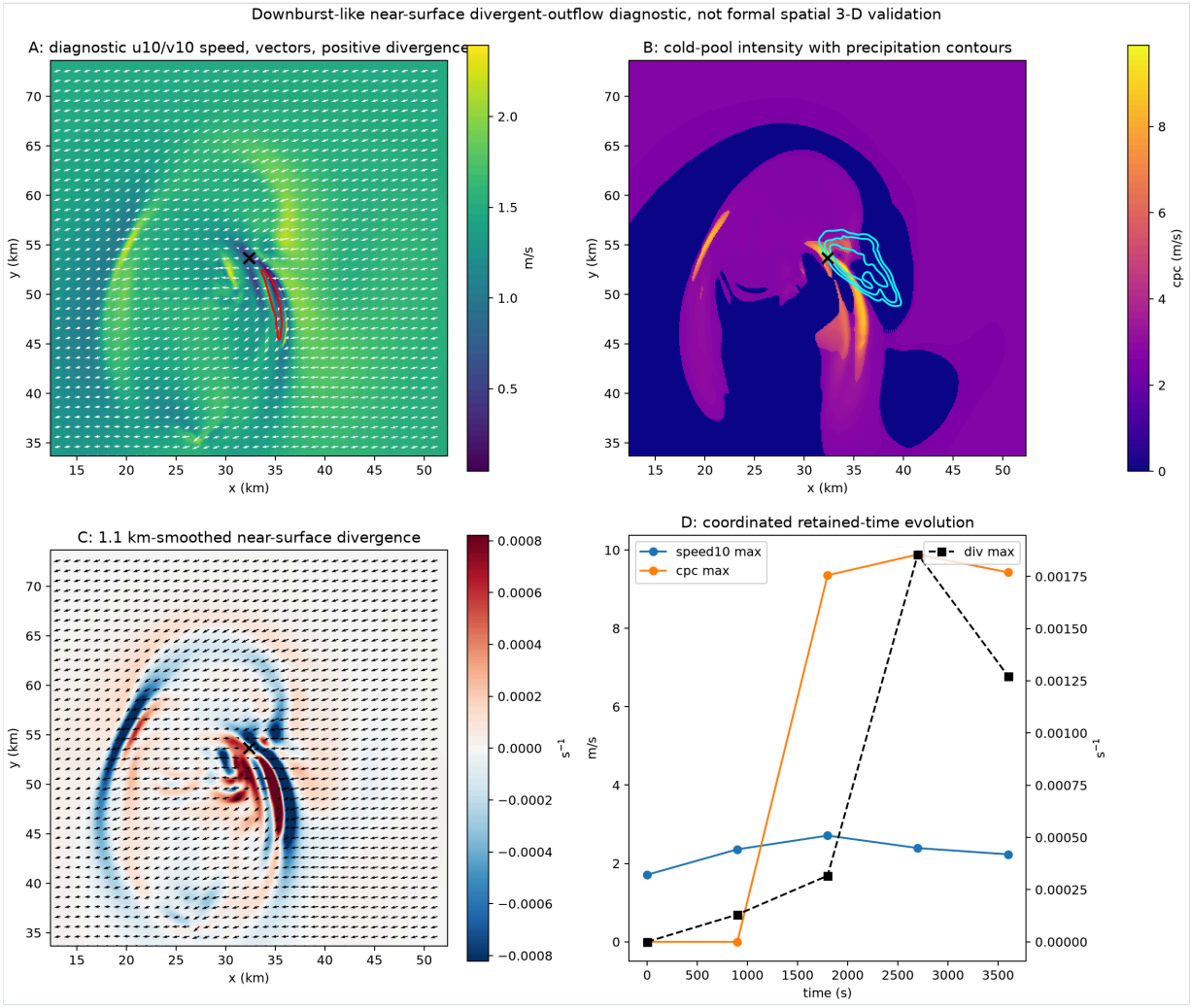
IF-15 **BESSIE V2 display, F03**



内容。 Discrete x-y positions of the main native-divergence center and the alternative smoothed surface-divergence center.
变量和单位。 Diagnostic center coordinates and center identity.
时间和坐标。 0, 900, 1800, 2700, and 3600 s; x-y positions.
证据类别。 `source-traceable display derivative`
来源。

claim limit。 Positions are discrete observations, not a continuous trajectory or event reconstruction.
源 SHA-256。 `ad2b5e5b2ef2872c09ac4ac23f93c7e01a17db33caca9a813eac6b241290d2f3`；**报告副本 SHA-256。** `ad2b5e5b2ef2872c09ac4ac23f93c7e01a17db33caca9a813eac6b241290d2f3`；**副本变换。** `none`

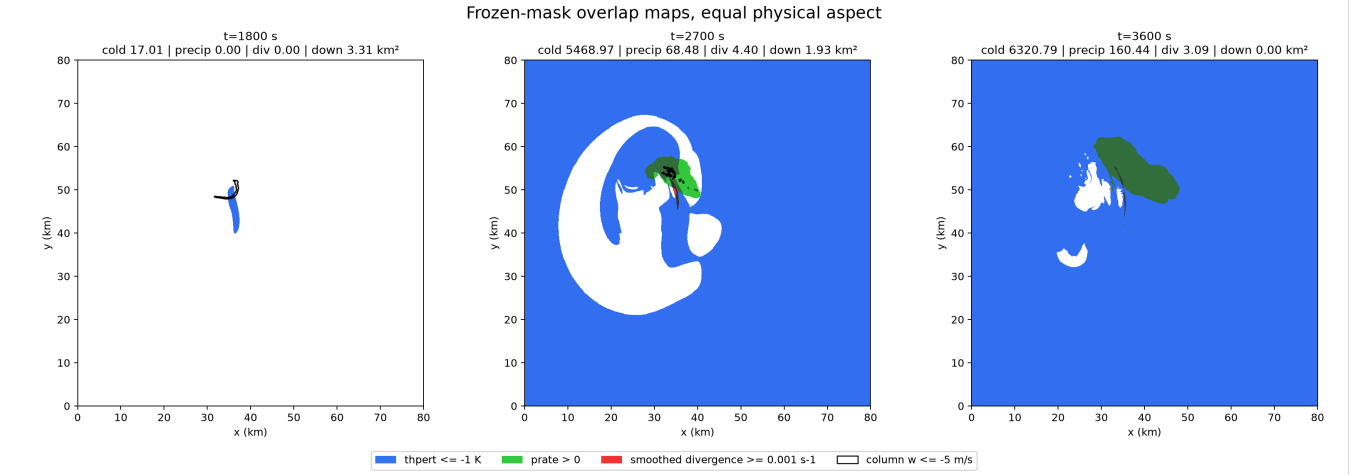
IF-16 **BESSIE V2 display, F04**



内容。 2700 s near-ground mechanism display with main and alternative center semantics.
变量和单位。 Near-ground source diagnostics, cold-pool and divergence-related fields.
时间和坐标。 2700 s; source spatial coordinates and units.
证据类别。 `source-traceable display derivative`
来源。

claim limit。 Supports model-internal display only and does not support a frozen-gate Level-1 promotion.
源 SHA-256。 `c67644738a70d26054c38f6522736a7e189101bc4dd9a05b36ec701b502b7957`；**报告副本 SHA-256。** `c67644738a70d26054c38f6522736a7e189101bc4dd9a05b36ec701b502b7957`；**副本变换。** `none`

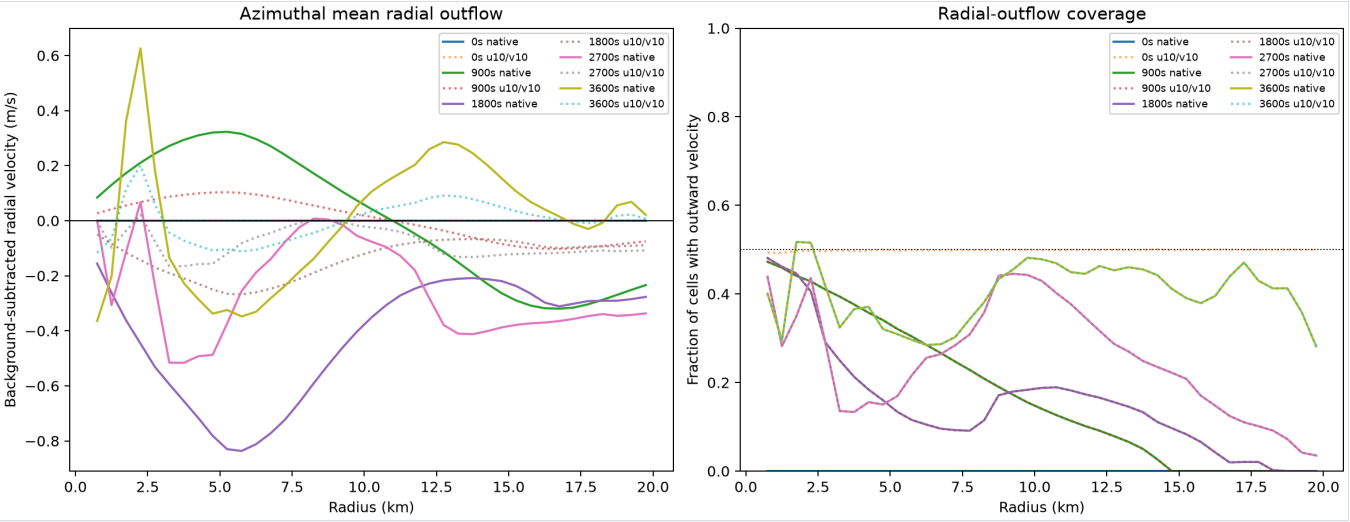
IF-17 **BESSIE V2 display, F05**



内容。 Overlap masks at 1800, 2700, and 3600 s with source-defined degraded semantics.
变量和单位。 Overlap masks and source diagnostic categories.
时间和坐标。 1800, 2700, and 3600 s; source grid.
证据类别。 `source-traceable display derivative`
来源。

claim limit。 Mask overlap is not event identity, observational truth, or formal downburst-validation evidence.
源 SHA-256。 `bf7d11bd9d6060e75b0bdc7783363f820b46e59aefc17302a69b59017726bcfc`；**报告副本 SHA-256。** `bf7d11bd9d6060e75b0bdc7783363f820b46e59aefc17302a69b59017726bcfc`；**副本变换。** `none`

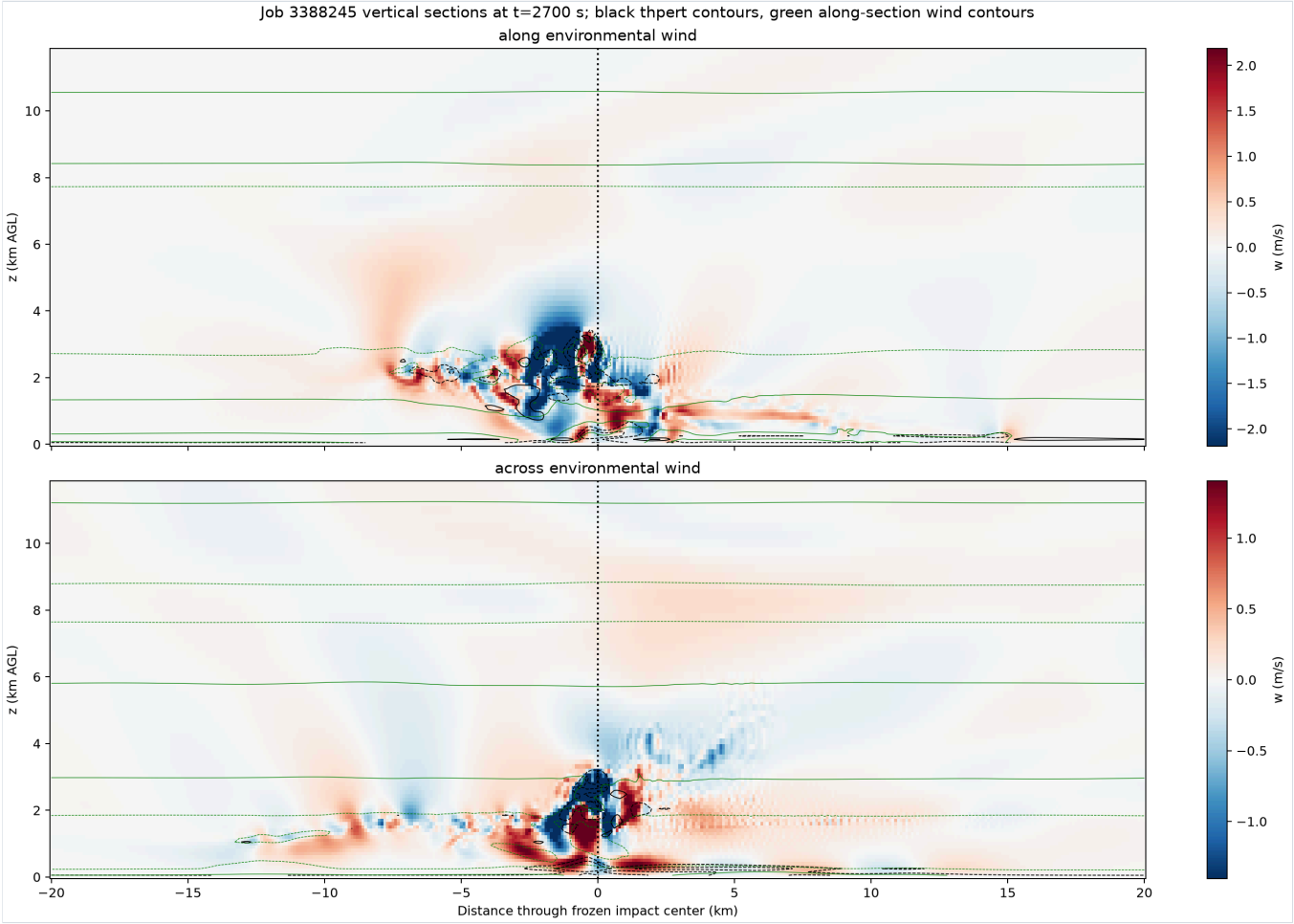
IF-18 **BESSIE V2 display, F06**



内容。 Radial and azimuthal outflow diagnostics at 2700 s.
变量和单位。 Radial component and azimuthal diagnostics; m/s where shown.
时间和坐标。 2700 s; radial and azimuthal support defined by the source package.
证据类别。 `source-traceable display derivative`
来源。

claim limit。 Radial diagnostic coverage does not establish VALID_DOWNBURST_CANDIDATE_LEVEL1 under the frozen gate.
源 SHA-256。 `1539841e0da38aec7e72efac1754dbd2ea896c253049287ae6686870c905f00d`；**报告副本 SHA-256。** `1539841e0da38aec7e72efac1754dbd2ea896c253049287ae6686870c905f00d`；**副本变换。** `none`

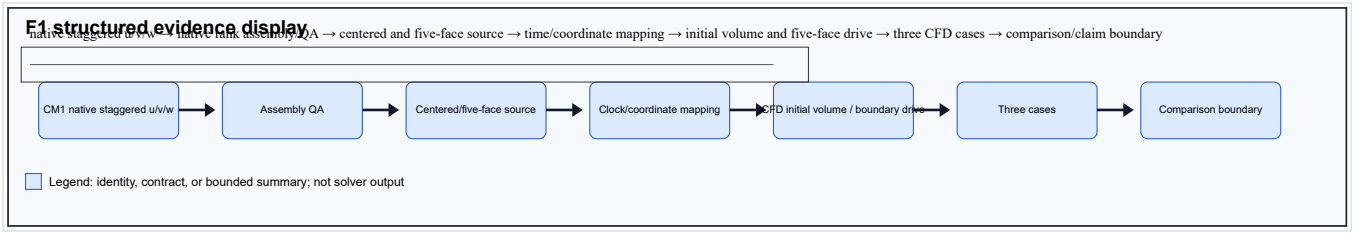
IF-19 **BESSIE V2 display, F07**



内容。 2700 s vertical sections emphasizing 0 to 3 km AGL.
变量和单位。 Vertical sections, downdraft, divergence and related source fields.
时间和坐标。 2700 s; height in m AGL where shown.
证据类别。 `source-traceable display derivative`
来源。

claim limit。 Vertical sections are conditional model fields and do not establish formal downburst validation or Vhat_3s.
源 SHA-256。 `b42a96948097343392550b34ab8bc760d2e53c4c8f4b5e237a17211db843b96b`；**报告副本 SHA-256。** `b42a96948097343392550b34ab8bc760d2e53c4c8f4b5e237a17211db843b96b`；**副本变换。** `none`

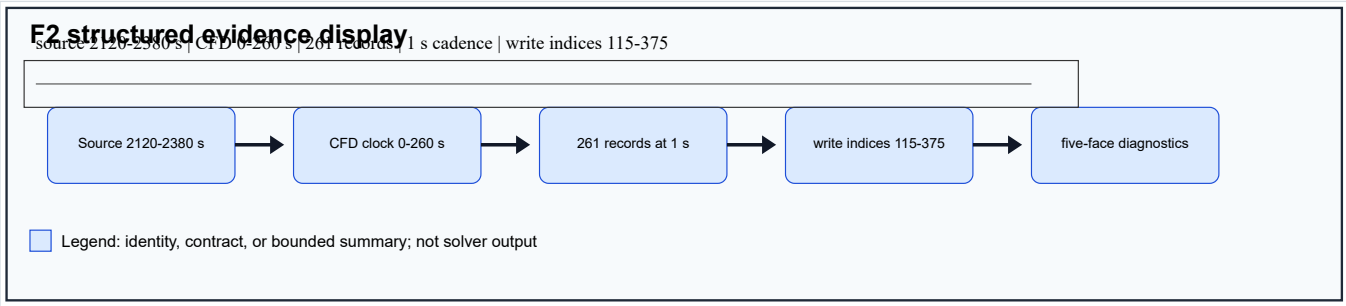
IF-20 **D420 V12 display, F1**



内容。 CM1 native staggered u/v/w to assembly, centered/five-face source, cases and comparison boundary
变量和单位。 u/v/w; m/s in mapping label
时间和坐标。 process diagram
证据类别。 `interface/design display derivative`
来源。

claim limit。 not solver output
源 SHA-256。 `8d2ea2aef567ae14b8ea0e0e32abb53e6596358718160d92e5122d8916119e4c`；**报告副本 SHA-256。** `8d2ea2aef567ae14b8ea0e0e32abb53e6596358718160d92e5122d8916119e4c`；**副本变换。** `none`

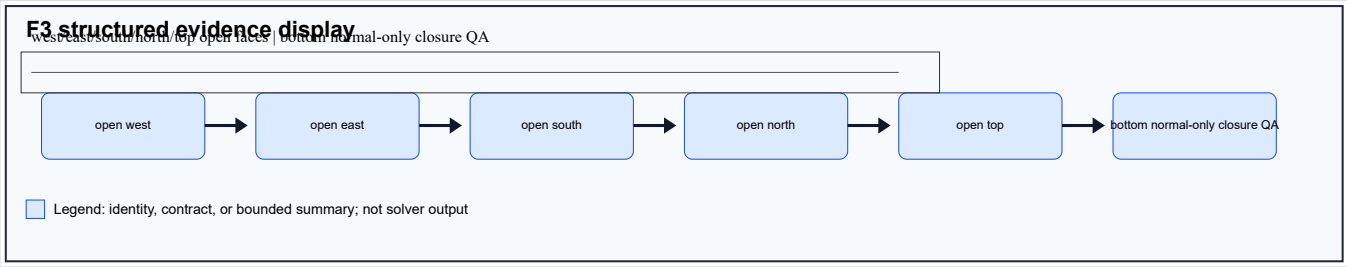
IF-21 **D420 V12 display, F2**



内容。 source and CFD clocks, records, cadence, indices, five-face diagnostics
变量和单位。 metadata; 261 records; 1 s
时间和坐标。 2120-2380 s to 0-260 s
证据类别。 `interface/design display derivative`
来源。

claim limit。 not solver output
源 SHA-256。 `117221ea2af5453a6cd5c429a591bbfdabfb4b2bdd3b612e08b5f1f6cd7a1a47`；**报告副本 SHA-256。** `2401222bd022c8d4c84a318b62c00b13a2113eb5dc941293e28a04db0aa3b3e3`；**副本变换。** `typographic_dash_normalization_only`

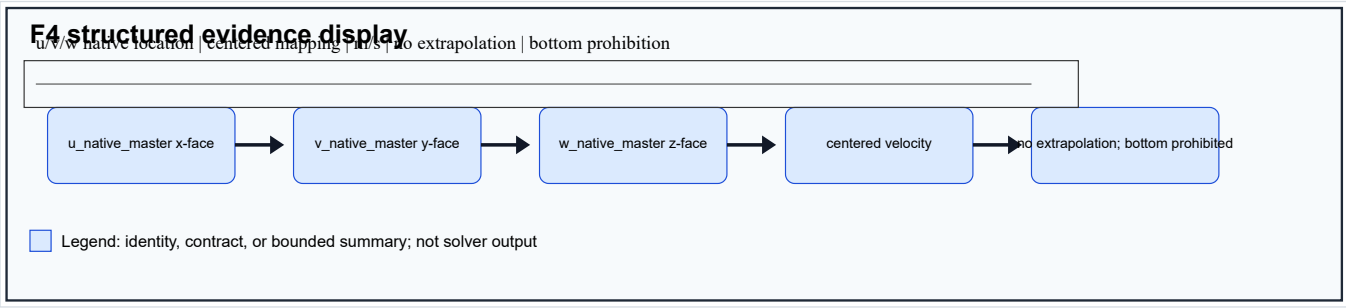
IF-22 **D420 V12 display, F3**



内容。 west/east/south/north/top open faces and bottom normal-only closure QA
变量和单位。 face contract
时间和坐标。 protocol diagram
证据类别。 `interface/design display derivative`
来源。

claim limit。 not solver output
源 SHA-256。 `988f3ef55387038e9303341531a70ef7bbe47e06f56e05c7ab3b6c78e58f96b1`；**报告副本 SHA-256。** `988f3ef55387038e9303341531a70ef7bbe47e06f56e05c7ab3b6c78e58f96b1`；**副本变换。** `none`

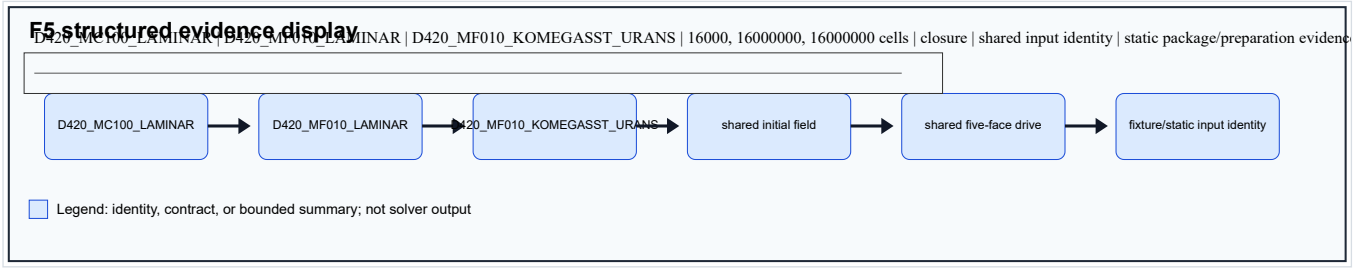
IF-23 **D420 V12 display, F4**



内容。 native u/v/w locations and centered mapping
变量和单位。 u/v/w; m/s
时间和坐标。 protocol diagram
证据类别。 `interface/design display derivative`
来源。

claim limit。 not solver output
源 SHA-256。 `9563bfe7274725903ea116fa6aa3c6ed20124a9bcb010bef51d1b198c7217395`；**报告副本 SHA-256。** `9563bfe7274725903ea116fa6aa3c6ed20124a9bcb010bef51d1b198c7217395`；**副本变换。** `none`

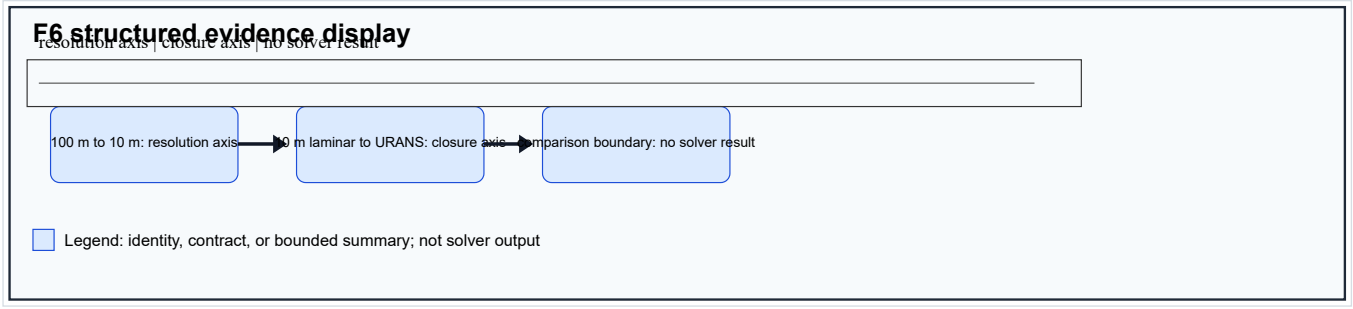
IF-24 **D420 V12 display, F5**



内容。 three static case identities and cell counts
变量和单位。 laminar/kOmegaSST; cells
时间和坐标。 100 m/10 m design identity
证据类别。 `interface/design display derivative`
来源。

claim limit。 not solver output
源 SHA-256。 `7dec98f9c0d9ac64f653a5301208160ac91221d4645b18bcb249dcd6a39a1aae`；**报告副本 SHA-256。** `7dec98f9c0d9ac64f653a5301208160ac91221d4645b18bcb249dcd6a39a1aae`；**副本变换。** `none`

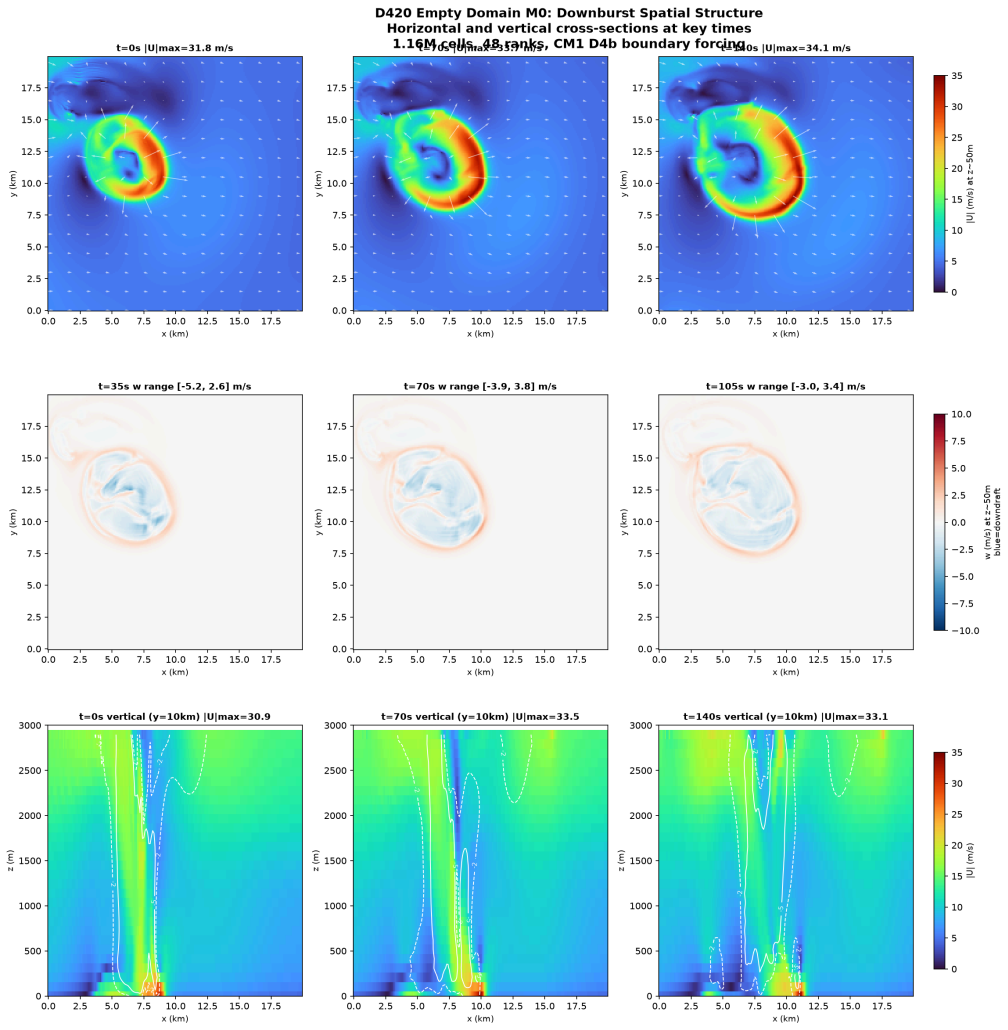
IF-25 **D420 V12 display, F6**



内容。 resolution and closure comparison axes
变量和单位。 not a solver variable
时间和坐标。 future design diagram
证据类别。 `interface/design display derivative`
来源。

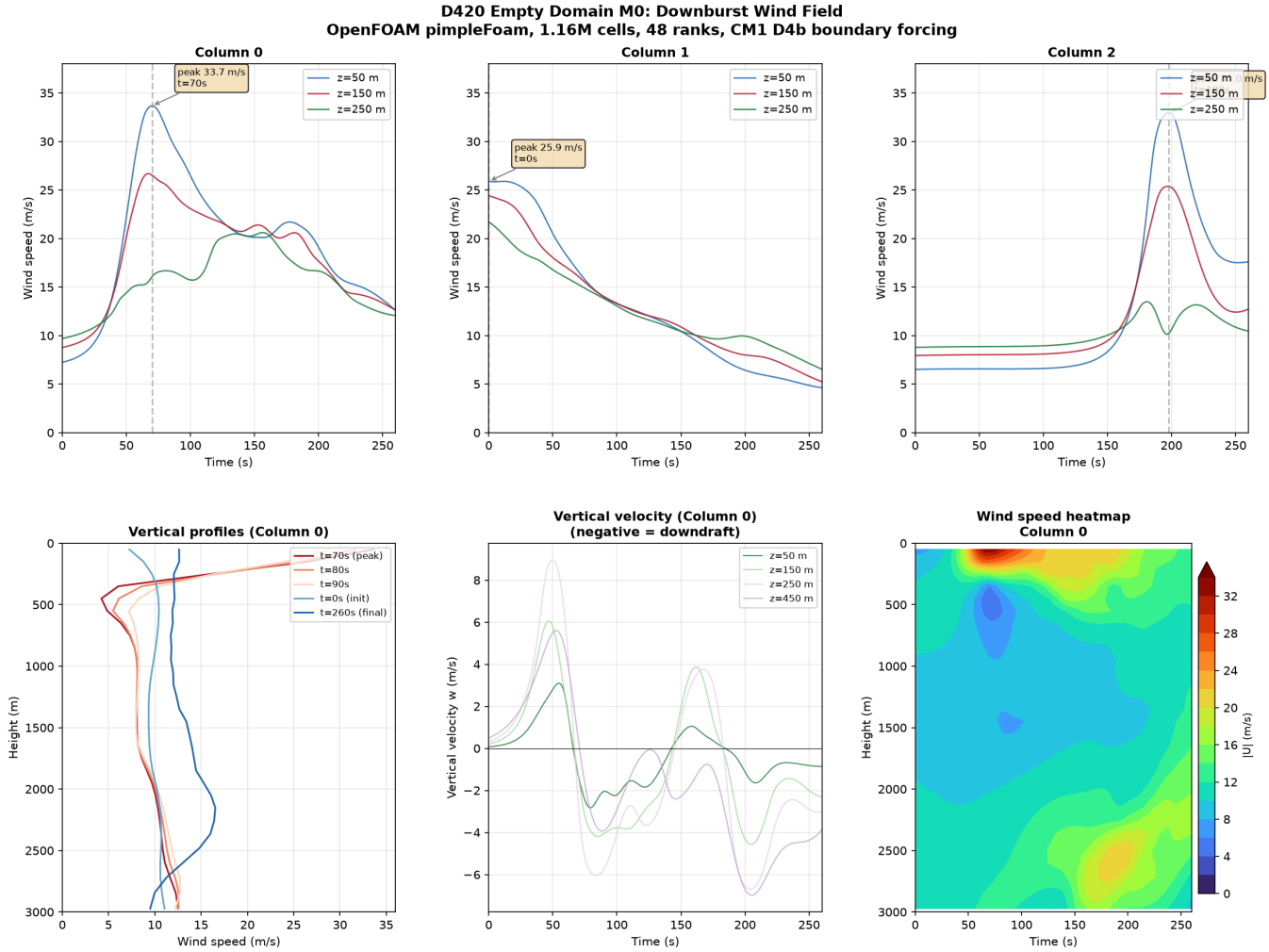
claim limit。 not solver output
源 SHA-256。 `920436f35e08166924059c02c3abee2af9502f3f795a6eb15f9a42b121192c95`；**报告副本 SHA-256。** `920436f35e08166924059c02c3abee2af9502f3f795a6eb15f9a42b121192c95`；**副本变换。** `none`

IF-26 **D420 V12 display, F7**



内容。 three rows: horizontal |U| with vectors; horizontal w; y=10 km vertical profiles
变量和单位。 |U| and w; m/s; z=50 m for top/middle; |U| colorbar in bottom
时间和坐标。 t=0/35/70/105/140 s; x/y km and x/z m
证据类别。 `M0 solver-derived postprocessing; generation-chain replay incomplete`
来源。
claim limit。 not controlled validation or design wind
源 SHA-256。 `f3e559db748bf9a00c1b711de4a9e9e397a3f59355967dfc80da56e0d332e514`；**报告副本 SHA-256。** `f3e559db748bf9a00c1b711de4a9e9e397a3f59355967dfc80da56e0d332e514`；**副本变换。** `none`

IF-27 **D420 V12 display, F8**



内容。 three column speed time series; Column 0 vertical profiles; Column 0 w; Column 0 heatmap
变量和单位。 wind speed, w, |U|; m/s; z=50/150/250/450 m
时间和坐标。 0-260 s; time/height axes
证据类别。 `M0 solver-derived postprocessing; generation-chain replay incomplete`
来源。
claim limit。 Column 2 peak annotation partially occluded; not validated gust
源 SHA-256。 `85247701bb9d3a625ba448f16ab8c7685ea8e622f908420ba45829e489a94d9b`；**报告副本 SHA-256。** `85247701bb9d3a625ba448f16ab8c7685ea8e622f908420ba45829e489a94d9b`；**副本变换。** `none`

7. Integrated tables

本报告把 component report 的关键表格、D420 T1-T9 table family 和 BESSIE diagnostic JSON sidecars 集成到 report-local `tables/`。全文表格可在 CSV/JSON 中读取，以下 HTML 表格展示最重要的 rows 并保留完整文件。

7.1 Input and output contracts

field	value	unit_or_semantics
protocol	PETW_MOORE_UNIFORM_AGL_V1	identifier
CM1	r20.3	version
grid	800 x 800 x 84	cells
duration	7200	s
cadence	60	s
output_filetype	3	CM1 output type
MPI	48	ranks
decomposition	6 x 8	MPI layout

field	value	unit_or_semantics
sws	cumulative lowest-model-level envelope	m/s
u10_v10	diagnostic near-surface wind	m/s, not validated 10 m 3 s gust
Vhat_3s	NOT_ESTABLISHED	not available

7.2 40-case cross-tab

level1_label	sws_max_ge_26_mps	sws_max_lt_26_mps	total
VALID_DOWNBURST_CANDIDATE_LEVEL1	10	14	24
WEAK_NEAR_SURFACE_OUTFLOW	1	8	9
COMPLETED_BUT_SCIENTIFICALLY_INVALID	1	2	3
BOUNDARY_TRUNCATED	2	0	2
STRONG_OUTFLOW_CANDIDATE_NEEDS_3D	1	1	2

7.3 Downstream claim boundary

use	status	boundary
Level-1 screening	SUPPORTED	model-internal 2-D evidence
CFD handoff		source semantics first
design wind/load	NOT_PROMOTED	not established

7.4 D420 execution and case identity tables

record	job_or_path	state	terminal_or_QA	actual_output	evidence_class	claim_boundary
CM1 source production	3379790	COMPLETED, 0:0	1026/1026 manifest; 20/20 cleanroom; 261/261 assembly; 261/261 frame-flux	CM1 source product	CM1 scientific/source	not CFD result
D420 M0	3384063	terminal markers complete; m0_rc=0	M0_DECOMPOSED; M0_GATE_PASS; D420_D7_V13_COMPLETE	M0 solver execution and F7/F8 derivatives	CFD execution + display	not validation or design guidance
D420 M1	3386652	COMPLETED, 0:0; solver rc=0	U,p,k,omega,nut at 50/60/70/80; 240 nonzero field checks	M1 restart execution	CFD execution	no superiority or coupling validation
M0/M1 postprocess	3393320	COMPLETED_PENDING_	reconstructPar, 8 NPZ, 4 metrics JSON, 4 figure inputs	bounded exploratory comparison	postprocessing/interpreted	NOT_CONSISTENT_CONTROLLED_COMPARISON
failed pre-reconstruction package	3389204	FAILED, 2:0	immutable package validator rejected .submit_once.lock	no scientific output	stopped execution	no result claim
failed rebound postprocess	3391286	FAILED, 2:0	processor mesh points absent in 48/48 processor dirs	no M1/M0 metric or figure result	stopped execution	no scientific result

case_id	job_or_state	mesh	cell_count	domain	closure	wall	boundary_or_initial	time	outputs	evidence_class	comparability
D420_M0_LAMINAR	3384063 completed	200×200×29; 100 m horizontal	1160000	20 km×20 km×3.038 km	laminar	ground slip	nonuniform U; mapped forcing; fallback uniform zero	0-260 s; write 1 s	U,p; F7/F8 |U| and w derivatives	actual solver + derived display	not controlled against M1
D420_M1_URANS	3386652 completed	400×400×100; 10 m	16000000	4 km×4 km×1 km	kOmegaSST URANS	ground noSlip + wall functions	restart from t=40; original 0/U uniform zero; different fallback	40.121951-80 s; selected 50/60/70/80	U,p,k,omega, nut	actual solver execution + field QA	not controlled against M0
D420_MC100_LAMINAR	package only	40×40×10; 100 m	16000	4 km×4 km×1 km	laminar	package-defined; not executed	shared source/drive identity in static package	not executed	input package only	generated preparation	future resolution arm
D420_MF010_LAMINAR	package only	400×400×100; 10 m	16000000	4 km×4 km×1 km	laminar	package-defined; not executed	shared source/drive identity in static package	not executed	input package only	generated preparation	future resolution arm
D420_MF010_KOMEGASST_URANS	package only	400×400×100; 10 m	16000000	4 km×4 km×1 km	kOmegaSST URANS	package-defined; not executed	shared source/drive identity in static package	not executed	input package only	generated preparation	future closure arm

7.5 Table inventory

id	title	relative_path	semantics	claim_limit
T01	CM1 V4 40-case distribution	tables/cm1_v4_TABLE_40_DISTRIBUTION.csv	Registry-derived counts: 24, 9, 3, 2, 2.	model-internal 2-D Level-1 screening only
T02	CM1 V4 40-case proxy cross-tab	tables/cm1_v4_TABLE_40_CROSSTAB.csv	Level-1 labels split by sws_max >= 26 m/s proxy.	Proxy is not Vhat_3s or validated 10 m 3 s gust.
T03	CM1 V4 full 40-case registry	tables/cm1_v4_TABLE_40_FULL_CASE_REGISTRY.csv	Full existing registry rows and case-level statuses.	No paper membership or formal 3-D validation.
T04	CM1 V4 all-102 identity status	tables/cm1_v4_TABLE_ALL102_IDENTITY_STATUS.csv	102 identity and production-status rows.	Input identity is not scientific completion.
T05	CM1 V4 all-102 status summary	tables/cm1_v4_TABLE_ALL102_STATUS.json	102, 40, 62, 61, 1 and Job 3396402 status fields.	Job is operational-only; 102 scientific results do not exist.
T06	CM1 input contract	tables/cm1_v4_TABLE_BENCHMARK_INPUT_CONTRACT.csv	Protocol, grid, duration, cadence, output type, MPI and decomposition.	Frozen protocol does not establish exact paper replication.
T07	CM1 output contract	tables/cm1_v4_TABLE_BENCHMARK_OUTPUT_CONTRACT.csv	sws and diagnostic near-surface variable semantics.	u10/v10 are not validated 10 m 3 s gust; Vhat_3s unavailable.
T08	CM1 process timeline	tables/cm1_v4_TABLE_BENCHMARK_PROCESS_TIMELINE.csv	Input, reconstruction and exact-reproduction status.	Process evidence is not scientific validation.
T09	CM1 downstream claim boundary	tables/cm1_v4_TABLE_DOWNSTREAM_USE_CLAIM_BOUNDARY.csv	Level-1 screening, CFD handoff and design wind/load status.	CFD and design uses remain gated.
T10	D420 CM1 source provenance	tables/d420_T1_CM1_SOURCE_PROVENANCE.csv	Source job, one-frame pilot, M0 chart, M1 QA and comparison audit provenance.	Source product and display derivatives are not one controlled result.
T11	D420 execution QA	tables/d420_T2_CM1_EXECUTION_QA.csv	M0/M1, source production, postprocessing and failed-job states.	Failed jobs have no scientific result.
T12	D420 variable semantics	tables/d420_T3_CM1_VARIABLE_AND_UNIT_SEMANTICS.csv	CM1 and CFD variable/support/units mapping.	Mapping semantics do not prove propagation validation.
T13	D420 clock and coordinate mapping	tables/d420_T4_CM1_TO_CFD_CLOCK_COORDINATE_MAPPING.csv	2120 to 2380 s source clock and CFD relative clock.	Historical M0/M1 did not use a proven common support.
T14	D420 boundary and flux contract	tables/d420_T5_BOUNDARY_FACE_AND_FLUX_CONTRACT.csv	Open faces, bottom, closure, domain, initial state and fallback differences.	Direct conflicts prevent single-factor attribution.
T15	D420 case identity and axes	tables/d420_T6_CFD_CASE_IDENTITY_AND_COMPARISON_AXES.csv	M0/M1 actual cases and future three-arm package.	Package-only rows are not solver results.
T16	D420 existing result metrics	tables/d420_T7_EXISTING_CFD_RESULT_STATUS_AND_METRICS.csv	M0 charts, M1 execution and bounded exploratory ranges.	NOT_CONSISTENT_CONTROLLED_COMPARISON is retained.
T17	D420 evidence classes	tables/d420_T8_EVIDENCE_CLASS_AND_CLAIM_BOUNDARY.csv	Source, pilot, solver, exploratory and preparation classes.	No scientific claim.
T18	D420 figure-to-source matrix	tables/d420_T9_FIGURE_TO_SOURCE_MATRIX.csv	Eight figure source paths, semantics and claim limits.	F7/F8 generation-chain replay remains incomplete.
T19	BESSIE temporal diagnostics	tables/bessie_TEMPORAL_DIAGNOSTICS.json	Source temporal diagnostics and extrema fields.	Model-internal sequence only.
T20	BESSIE spatial colocation metrics	tables/bessie_SPATIAL_COLOCATION_METRICS.json	Source spatial centers and colocated diagnostics.	No event reconstruction or Level-1 promotion.
T21	BESSIE radial outflow metrics	tables/bessie_RADIAL_OUTFLOW_METRICS.json	Source radial/azimuthal metrics and center sensitivity.	Radial diagnostics do not establish formal downburst validation evidence.

Full 40-case registry, all-102 identity table, D420 T1-T9 CSV/Markdown, and BESSIE diagnostic JSON are report-local copies of the existing source packages. No raw solver output was copied.

8. Future work and gates

后续工作必须在当前 evidence boundary 之外以独立、可审批、可回读的 packet 进行。该报告不启动下列工作。

future item	current gate	minimum next evidence	stop condition
All-102 production	61 release-ready new inputs; 1 source-height blocker	exact approval, fresh ComputeOps strategy, one gate package, one output root and bounded scheduler wave	source identity, package and terminal QA fail or source-height case remains unresolved
Level-2 dense-source selection	40-case evidence available; sufficiency remains decision	selection criteria, source semantics, boundary clearance, cadence and spatial support	selected case cannot satisfy CM1-to-CFD packet contract
CFD matched drive	D420 three-arm package is NOT_EXECUTED preparation	new package-specific approval, same source/drive identity, same domain, wall, closure, initial state and boundary history	any mismatch prevents single-factor attribution
Windfield/EVA handoff	diagnostic manifests only	semantic closure for variables, cadence, coordinates, finite values and claim boundary	diagnostic field would be used as validated engineering gust
High-rise review English counterpart	ARCHIVED_CANDIDATE_BLOCKED	restore exact required model route and rerun bounded counterpart workflow	required model identity or source semantics cannot be verified

9. Claim boundary and stopping condition

以下 labels are preserved exactly where they occur in the source packages: `NOT_ESTABLISHED`, `NOT_SUPPORTED`, `NOT_PROMOTED`, `PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION`, , `not_scientifically_approved`, `SOURCE_HEIGHT_ORDER_BLOCKED`, `NOT_SUPPORTED_AS_VALID_DOWNBURST_CANDIDATE_LEVEL1_UNDER_FROZEN_GATE`, `NOT_CONSISTENT_CONTROLLED_COMPARISON` and . These are evidence-status labels, not interchangeable scientific conclusions.

本报告不建立： exact Moore 97-case reproduction; exact paper membership; observational event truth; formal downburst validation; formal three-dimensional downburst validation; validated 10 m 3 s gust; `Vhat_3s`; design wind; design load; CFD solver validation; CM1-to-CFD propagation validation; mesh independence; closure validation; code guidance; publication permission; or a scientific claim.

Stopping condition is satisfied after one local report-production pass, source-copy verification, HTML rendering, PDF build when available, PDF text check, representative visual check, figure/table index read-back, and self-excluded hash-manifest replay. No successor simulation, review revision, remote action or claim promotion is initiated by this report.

10. Concluding remarks

CM1 now has a self-contained bot-level evidence view that links its automation role, two review families, two governed Wiki/RAG namespaces, the frozen Moore protocol, the 40-case Level-1 registry, the all-102 identity inventory, the BESSIE conditional branch and the D420 CM1-to-CFD branch. The integrated figure set shows protocol, input identity, model-internal diagnostics, status inventories, interface contracts, solver-derived display derivatives and future package design without treating them as one scientific result.

The strongest current CM1 result is a protocol-consistent reconstructed evidence layer with explicit identity, runtime, diagnostic and downstream-use boundaries. The record remains `ACCEPT_IMPROVED_NOT_EXACT` for the Moore reconstruction, `NOT_ESTABLISHED` for exact paper reproduction and formal 3-D validation, `NOT_SUPPORTED` for the frozen BESSIE Level-1 promotion, and `NOT_PROMOTED` for Path C, Vhat_3s and design use. remains the claim gate for further production, CFD selection, publication use and scientific interpretation.